

LAB MANUAL

FUNDAMENTALS OF ELECTRICAL AND ELECTRONICS LAB (20EE01P)

II SEM

NAME OF STUDENT :

REGISTER NUMBER :

ROLL NUMBER :

NAME OF THE DEPT. :

BATCH :

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

GOVERNMENT POLYTECHNIC, BELLARY

Academic year 2022-2023

UNIT-1**Date:****1. Electrical symbols related to electrical engineering.**

 Resistor IEC System	 Resistor NEMA System	 Inductor	 Capacitor	 Switch	 Changeover	 Push-Button	 Male Plug IEC System	 Male Plug NEMA System
 Female Plug IEC System	 Female Plug NEMA System	 Fuse	 Electric Line	 Ground	 Diode	 Diac	 Thyristor	 Triac
 IC Chip	 Amplifier	 Electric Generator	 Battery	 Transistor	 Vacuum tube	 AND Gate ANSI System	 OR Gate ANSI System	 NAND Gate ANSI System
 NOR Gate ANSI System	 NOT Gate ANSI System	 AND Gate IEC System	 OR Gate IEC System	 AND Gate NEMA System	 OR Gate NEMA System	 7 Segment Display	 Ammeter	 Voltmeter
 Ohmmeter	 Frequency Meter	 Wattmeter	 Electric Clock	 Electrical Counter	 Recording Instrument	 Antenna	 Speaker	 Microphone
 Light Bulb	 Direct Current	 Alternating Current	 Positive Polarity	 Negative Polarity	 Piezoelectric Crystal	 Relay	 Transformer	 Electric Motor

Various types of safety sign

Shape	Meaning	Color	Examples
 Circle with diagonal bar	Prohibition	RED (contrast: white)	No smoking 
 Circle	Mandatory Action	BLUE (contrast: white)	Wear Eye protection 
 Equilateral Triangle	Warning	YELLOW (contrast: black)	Danger Flammable material 
 Square / Rectangle	Information about safe condition	GREEN (contrast: white)	Escape Route – Left 
 Square / Rectangle	Fire Safety	RED (contrast: white)	Fire Extinguisher 

2. Electrical safety

Electrical safety is a general practice of workers who are exposed to handle and maintain electrically powered equipment. It is a set of guidelines they follow to mitigate electrical hazards and prevent its dangerous effects in case of an incident

Disconnect completely. Which means that the electrical installation must be disconnected from live parts on all poles.

1. Secure against connection. ...
2. Verify that the installation is dead. ...
3. Carry out earthing and short-circuiting. ...
4. Provide protection against adjacent live parts.
5. Prevent electrical equipment from contacting wet wires
6. Ensure safe when unplugging
7. Install properly and tidy electrical cords
8. Lookout for electrical lines

3. Electrical earthing

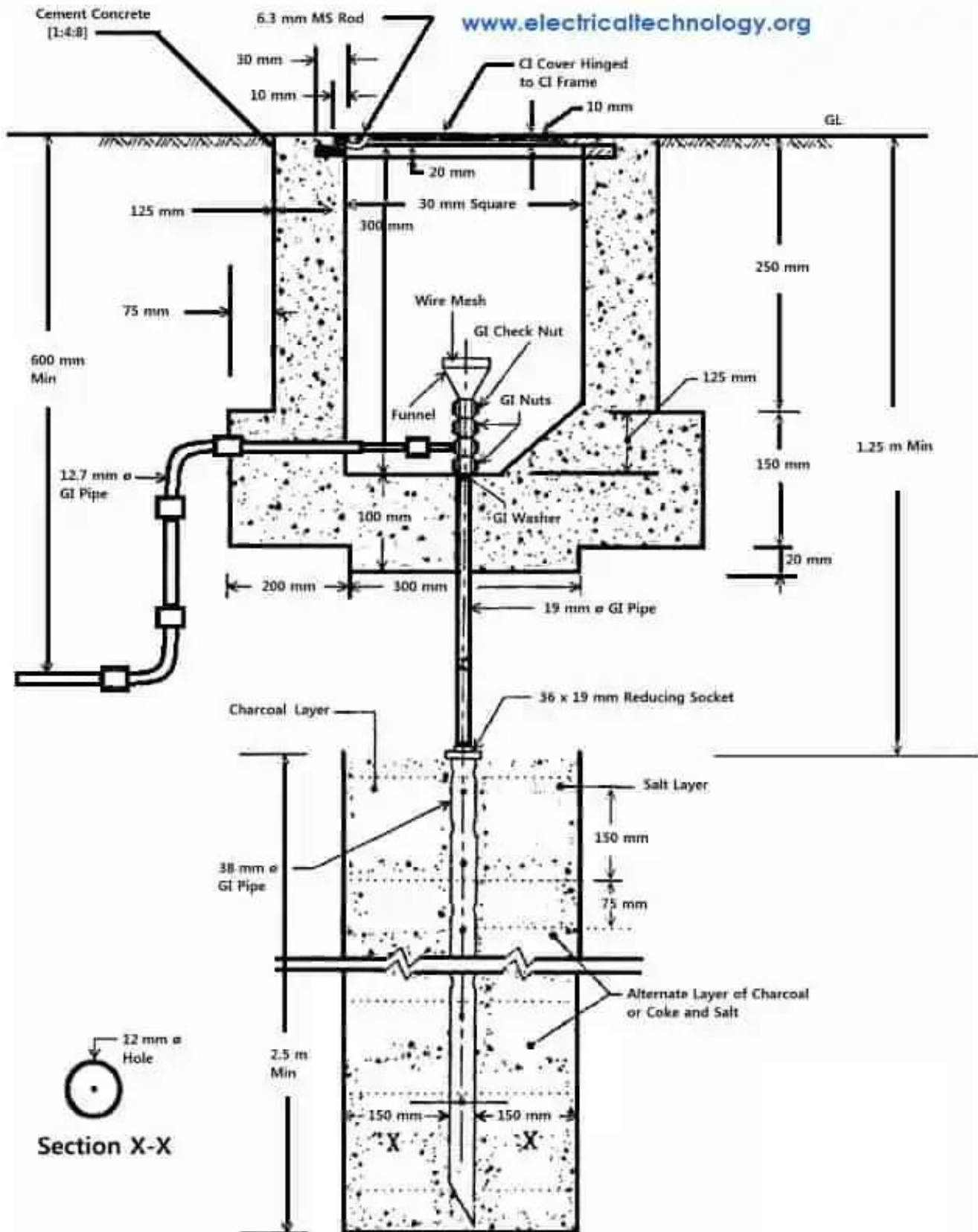
Definition: The process of transferring the **immediate discharge of the electrical energy directly to the earth by the help of the low resistance wire** is known as the electrical earthing. The electrical earthing is done by connecting the non-current carrying part of the equipment or neutral of supply system to the ground.

Types of Earthing

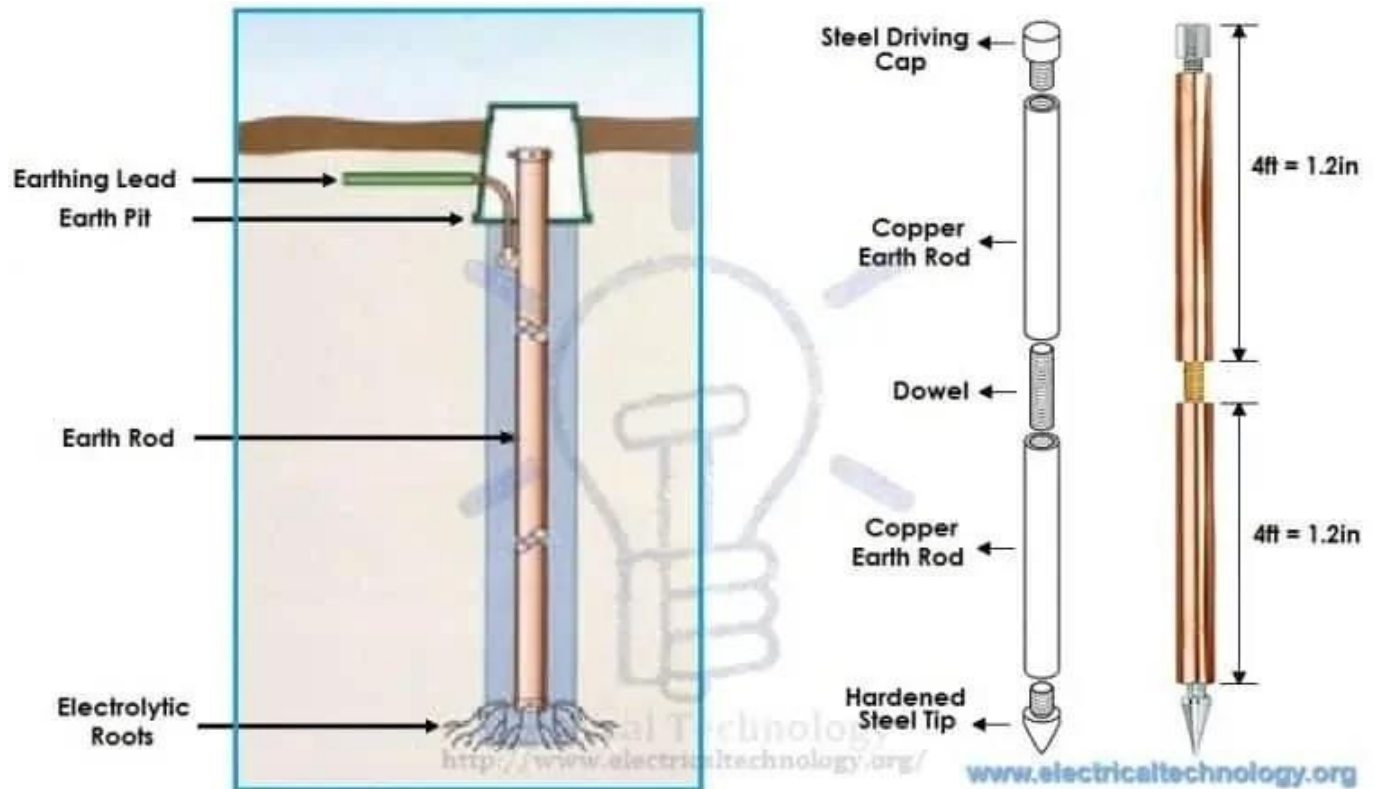
There are three types of earthing, they are:

1. Pipe earthing
2. Plate earthing
3. Rod earthing
4. Strip earthing

Pipe earthing is the best and most efficient way of earthing and is also easily affordable. Pipe earthing uses 38mm diameter and 2 metres length pipe vertically embedded in the ground to work as earth electrodes.



Pipe Earthing



Copper Rod Electrode Earthing System

Strip earthing is used in transmission processes. Strip electrodes of cross section not less than 25mm X 1.6mm of copper or 25 mm X 4mm of G.I. or steel are buried in horizontal trenches of a minimum depth of 0.5m.

Advantages of Earthing

1. Earthing is the safe and the best method of offering safety. We know that the earth's potential is zero and is treated as Neutral. Since low equipment is connected to earth using low resistance wire, balancing is achieved.
2. Metal can be used in electrical installations without looking for its conductivity, proper earthing ensures that metal does not transfer current.
3. A sudden surge in voltage or overload does not harm the device and person if proper earthing measures are done.
4. It prevents the risk of fire hazards that could otherwise be caused by the current leakage.

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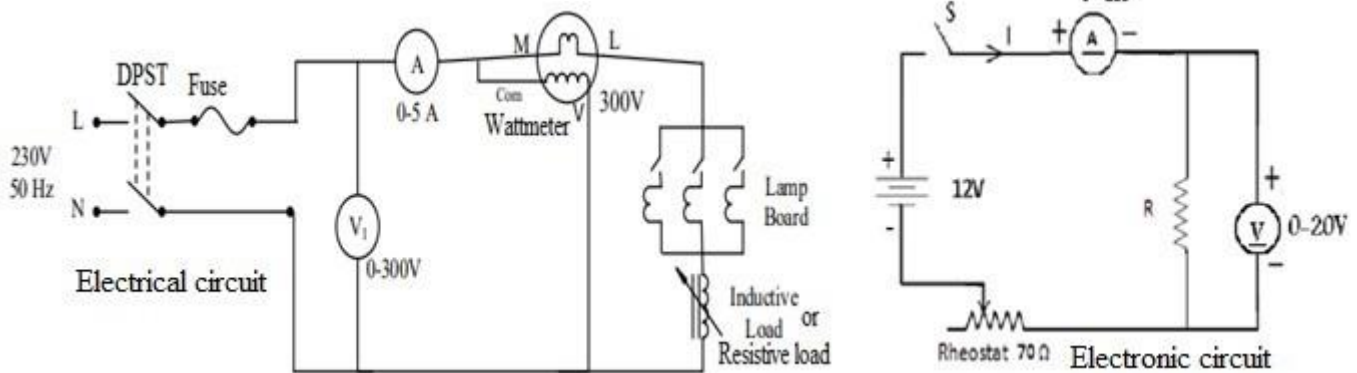
UNIT-2**Date:**

1. Connect voltmeter and ammeter in a simple circuit. (Practicing of identification and connection of different meters)

Aim: To Connect voltmeter and ammeter in a simple circuit. (Practicing of identification and connection of different meters.

Apparatus required

- | | |
|-------------------------------|---|
| 1. Ammeter 0-5A-1no | 1. Ammeter MC 0-1A (or DMM) - 01 No. |
| 2. Voltmeter 0-300v-1no | 2. Voltmeter MC 0-20 V(or DMM) - 01 No. |
| 3. Wattmeter 0-300v,0-5A -1no | 3. RPS-DC Supply 12V - 01 No. |
| 4. Connecting wires | 4. Rheostat - 50, 70, 75 Ohms - 01 No. |
| | 5. Wire wound resistor - - 01 No. |
| | 6. SPST (Knife) Switch - 01 No. |
| | 7. Connecting wires. |

Circuit diagram**Procedure:**

1. Use DPST switch, Ammeter, Voltmeter and Wattmeter.
2. Identify the meters that you will connect
3. Make Connections as per the circuit diagram.
4. Repeat the step 1&2 till you get good practice.

Result: voltmeter, ammeter, wattmeter are connected in a simple circuit for Practicing of identification and connection of different meters.

1. Measure current, voltage and analyze effective resistance in series circuit.

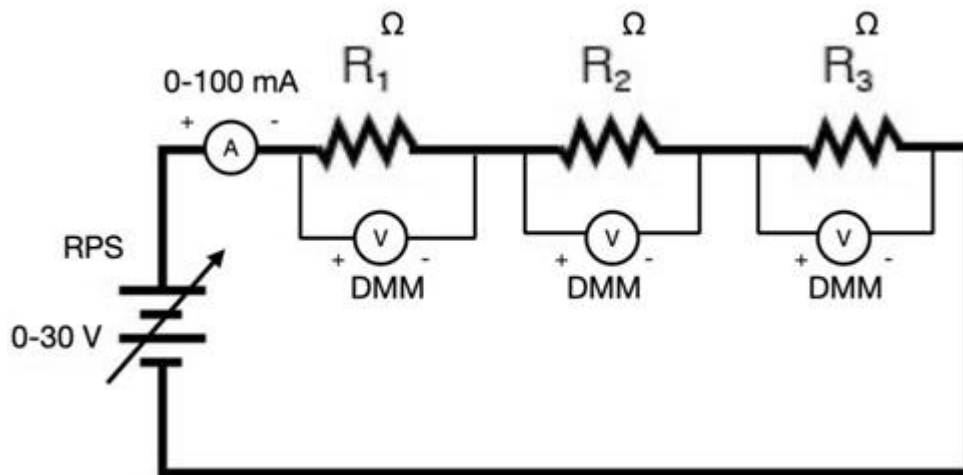
Date:

Aim: To Measure current, voltage and analyze effective resistance in series circuit.

Apparatus required

Sl.No.	Item with Specification	Quantity
1	Digital Multimeter	1
2	Ammeter DC, 0 to 100 mA	1
3	Resistors Ω , Ω , Ω	1 each
4	Regulated Power Supply, 0 to 30 V DC	1
5	Bread board	1
6	Connecting Wires	Few

CIRCUIT DIAGRAM:



PROCEDURE:

1. Check all the components for proper working.
2. Check the Resistance values by using Multimeter.
3. Rig up the circuit as shown in the circuit diagram.
4. Switch ON the Power supply.
5. Vary the applied voltage in steps of 2V
6. Note down the Voltmeter readings V1, V2 & V3 for each case.

7. Verify the equation for each case. i.e, $V=V_1+V_2+V_3$.

Res voltage in volts	Current 'I' in mA	Voltage V1 in Volts	Voltage V2 in Volts	Voltage V3 in Volts	$V=V_1+V_2+V_3$

THEORETICAL CALCULATION:

Let $R_1 = \text{_____} \Omega$, $R_2 = \text{_____} \Omega$ & $R_3 = \text{_____} \Omega$

Then $R_{\text{eff}} = R_1 + R_2 + R_3 = \text{_____} \Omega$

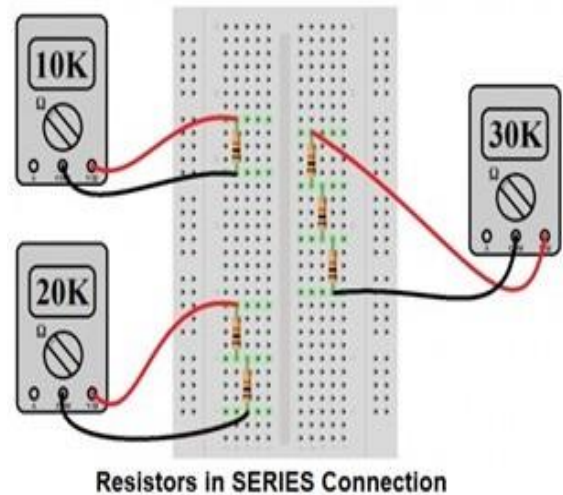
Total current $I = V / R_{\text{eff}}$.

$V_1 = I * R_1 = \text{-----} V$

$V_2 = I * R_2 = \text{-----} V$

$V_3 = I * R_3 = \text{-----} V$

$V = V_1 + V_2 + V_3$ (Volts)



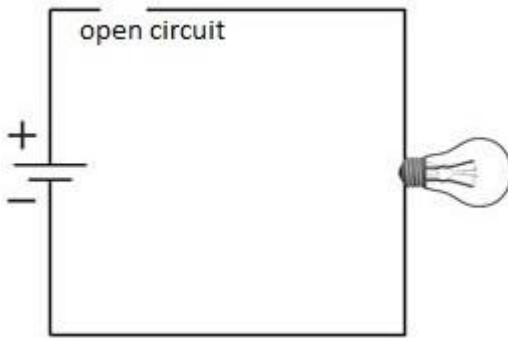
RESULT: Measurement of current, voltage and effective resistance in series circuit is calculated

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2. Demonstrate effects of shorts and opens in a circuit

Open circuit: An open circuit is a circuit where the path has been interrupted or "opened" at some point so that current will not flow. An open circuit is also called an incomplete circuit.

An open circuit is an electric circuit that does not allow current to pass. In a circuit, current can only flow if it finds a continuous path this is referred to as a "closed circuit." An open circuit if there is a break anywhere in the circuit, and current cannot flow



No current can flow if the circuit is broken, for example, when a switch is open in the circuit, current will not flow in any part of circuit as circuit is open and there is no closed path available for charge to flow.

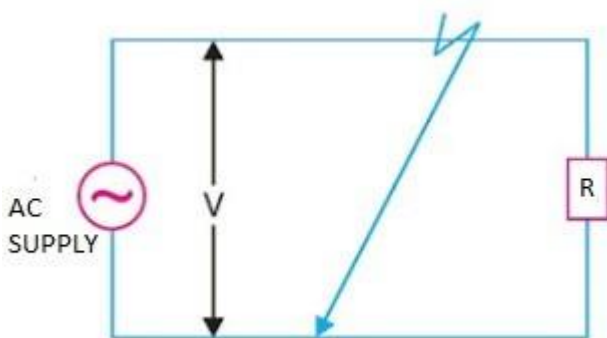
Short-Circuit

A short circuit is simply a low resistance connection between the two conductors supplying electrical power to any circuit. This results in excessive current flow in the power source through the 'short,' and may even cause the power source to be destroyed

Effects of Short-Circuit

When a short-circuit occurs, the current in the system increases to an abnormally high value while the system voltage decreases to a low value.

The **heavy current** due to short-circuit causes excessive heating which may result in **fire or explosion**. Sometimes short-circuit takes the form of an arc and causes considerable damage to the system



RESULT: effects of shorts and opens in a circuit is identified

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1. Determine the equivalent Resistance of parallel connected resistances.

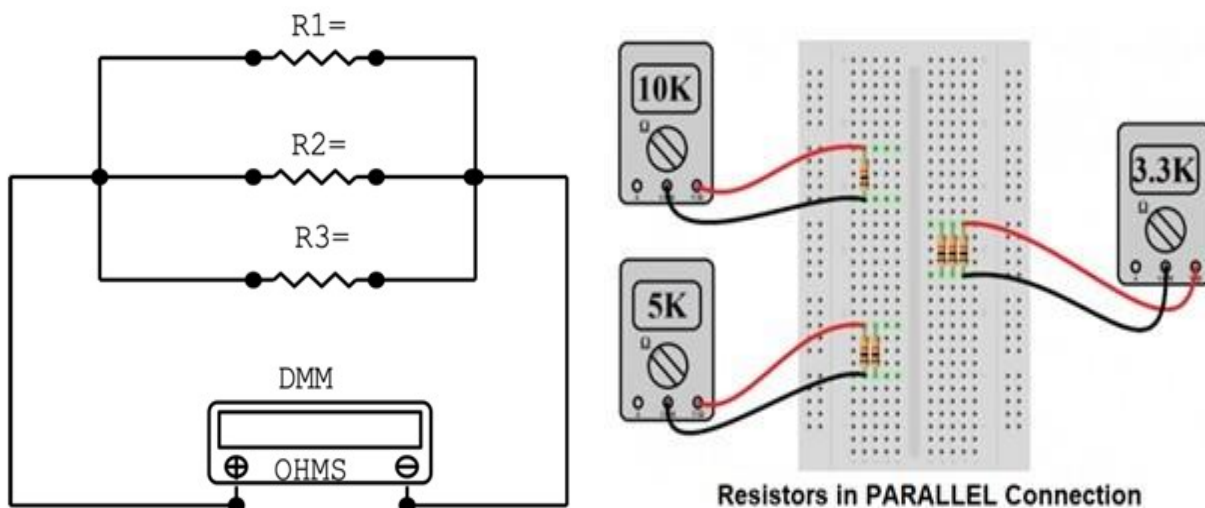
Date:

Aim: To determine the equivalent Resistance connected in parallel.

Apparatus required

SLNo.	Item with Specification	Quantity
1	Digital <u>Multimeter</u>	1
2	Resistors -	1 each
3	Bread board	1
4	Connecting Wires	Few

Circuit diagram



PROCEDURE:

1. Check the components (Resistors & Multimeter) for proper working.
2. Measure the Resistance values of all resistors by using Multimeter.
3. Rig up the circuit as shown in the circuit diagram.
4. Note down the Multimeter readings & tabulate the results. Let this value will be a practical value.
5. Compare this practical value with the theoretical value for each case.
6. Repeat the Experiment for different resistance values.

Tabular column

<u>Sl no</u>	R1	R2	R3	R_T=
<u>1</u>				
<u>2</u>				

CALCULATION:

$$1/R_T = 1/R_1 + 1/R_2 + 1/R_3$$

Practically, the Equivalent Resistance value is found to be, $R_{equ} = \underline{\hspace{2cm}} \Omega$

RESULT: It is found that the practical value of equivalent resistance when three Resistors are connected in parallel is approximately equal to the theoretical value. Hence the experiment is verified.

Signature of staff

1. Generate and demonstrate the measurement of frequency, time period and phase difference of AC quantity using CRO and function generator. Date:

THEORY on CRO:

1. CRO is a basic instrument employed for the study of several types of waveforms.
2. It can measure various quantities such as peak voltage, frequency, phase difference, pulse- width, delay time, rise time, and fall time.

DISPLAY CONTROLS

Common display control includes:

1. **An intensity control** is used to adjust the brightness of the waveform. As the sweep speed is increased, there is a need to increase the intensity level.
2. **A focus control** is used to adjust the sharpness of the waveform.
3. **A trace control** is used to rotate the trace on the CRO screen.
4. **A calibration point** is used to calibrate the CRO. It gives a steady square wave at a particular set frequency and voltage. It allows the accurate scaling of the trace. The standard calibration signal is 0.2V at 1 KHz.

VERTICAL CONTROLS

Vertical controls are used to position and scale the wave form vertically.

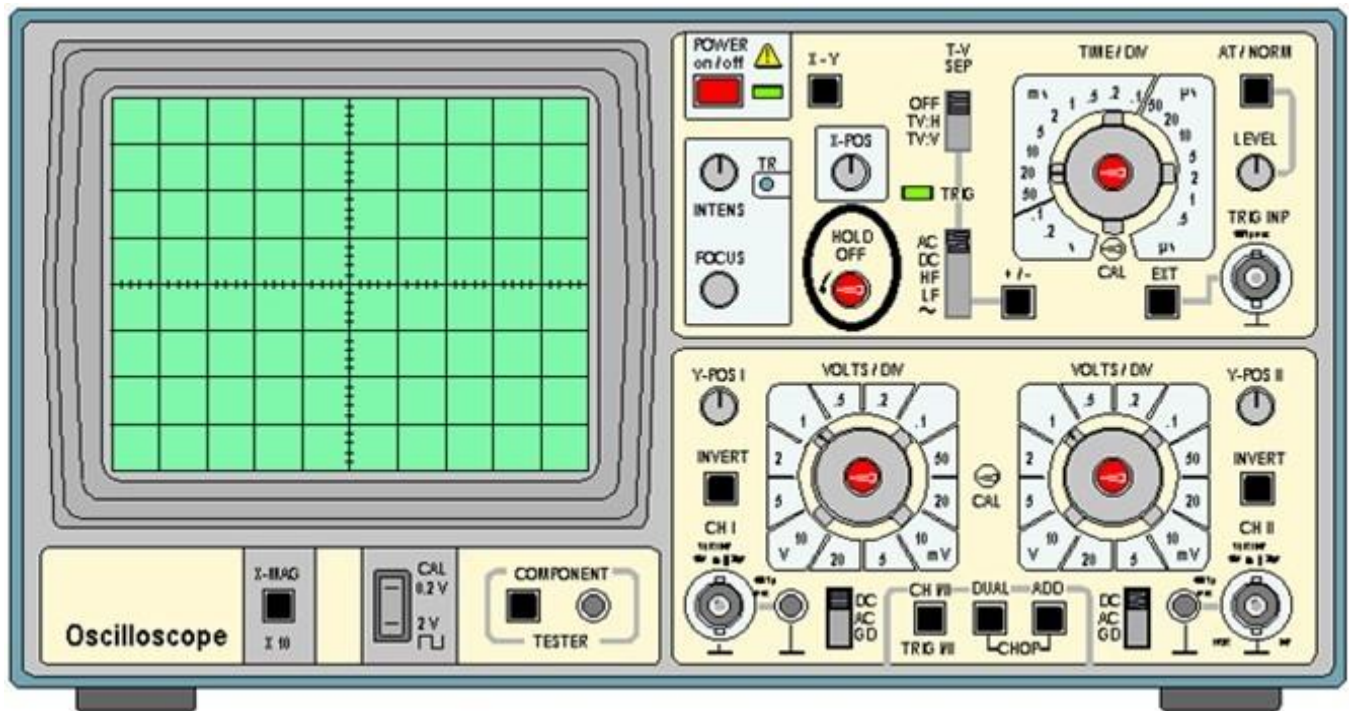
1. **Volts/div.:** For selecting desired voltage sensitivity of the vertical amplifier to obtain the proper wave forms on the screen.
2. **Vertical Position knob:** To move the trace up or down the on the screen.

HORIZONTAL CONTROLS

Horizontal controls are used to position and scale the wave form horizontally.

1. **Time base (time/div) :**For selecting desired sweep rate from calibrated steps or admits external signal to horizontal amplifier.

2. **X Position:** Controls horizontal position of trace onscreen.

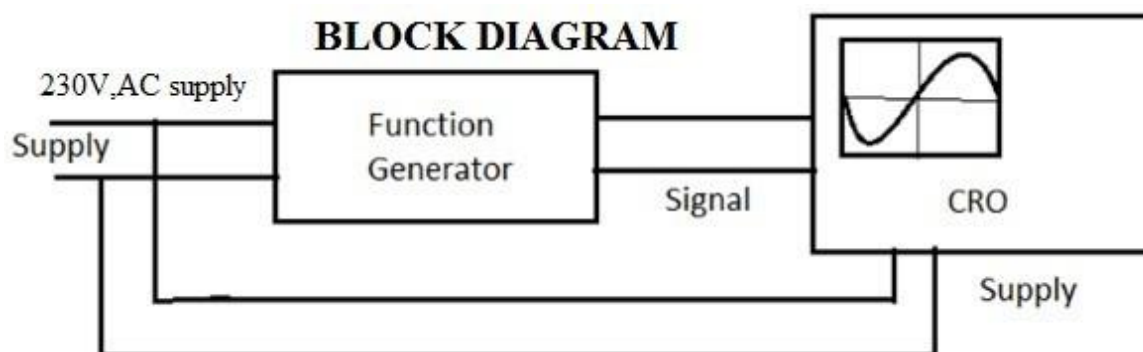


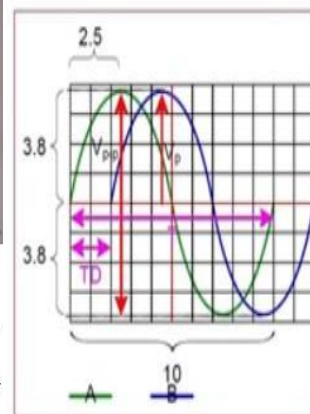
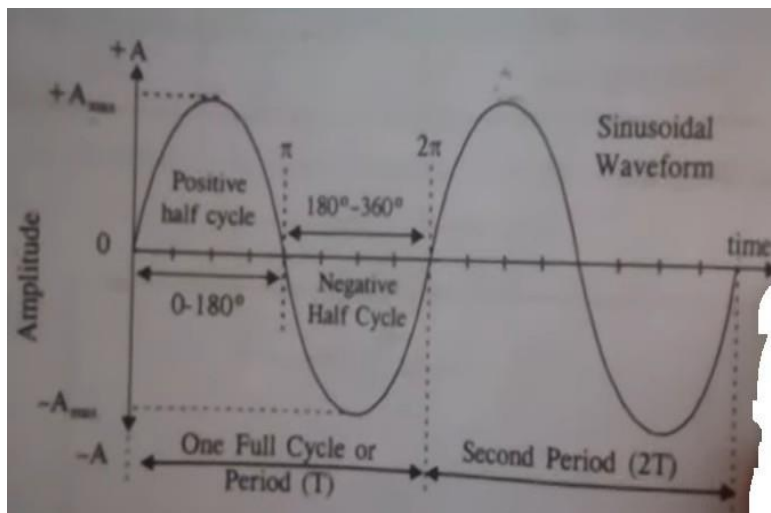
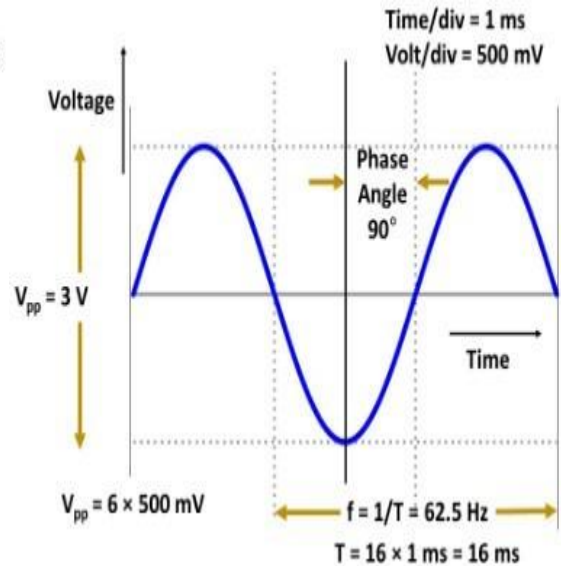
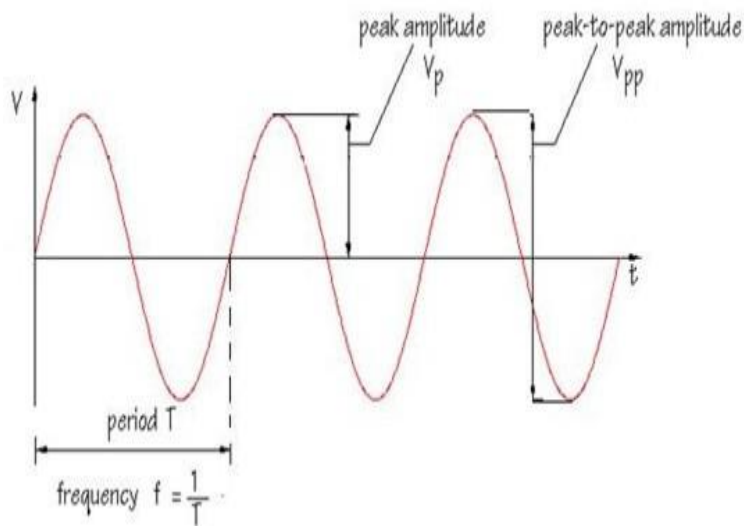
Aim: To measure the frequency, time period, peak voltage and peak to peak voltage of AC quantity using CRO and function generator

APPARATUS:

1. Signal generator-01,
2. Cathode Ray Oscilloscope (CRO)-01,
3. Function Generator
3. CRO Probes-02.

CIRCUIT DIAGRAM:





- a) Voltage Peak-to-Peak
 $V_{pp} = (V/\text{Div}) \times \text{No. of vert. div.}$
 $= 100 \text{ mV/div} \times (3.8 \times 2)$
 $= 0.76 \text{ V}$
- b) Voltage Peak
 $V_p = (V/\text{Div}) \times \text{No. of vert. div.}$
 $= 100 \text{ mV/div} \times (3.8)$
 $= 0.38 \text{ V}$

(Volt/Div : 100mV/Div, Time/Div : 0.5ms/Div)

Where V_p = peak VOLTAGE in volts
 V_{pp} = peak-to-peak VOLTAGE in volts
 T = Time period in seconds
 F = frequency in Hertz

PROCEDURE:

1. Connect the Signal generator to CRO by using CRO probes.
2. Switch ON the Power supply.
3. Display the AC voltage waveform on CRO screen.
4. Set the required voltage and frequency for the input waveform in signal generator.
5. Measure the Amplitude, Peak-to-Peak voltage, Frequency, Time Period and Wavelength as shown in the diagram.
6. Note down and tabulate the results.

TABULATION

sl no	no of horizontal division(X)	Time / division (y)	$T=X*Y$	$F=1 / T$	No. of Vertical Divisions(X)	Voltage/ Division (Y)	$V_{p-p}=X*Y$

CALCULATION

Time Period (T) = no of horizontal division x Time per division at knob in ms

=

= milli seconds

Frequency (f) = $1/T$ =

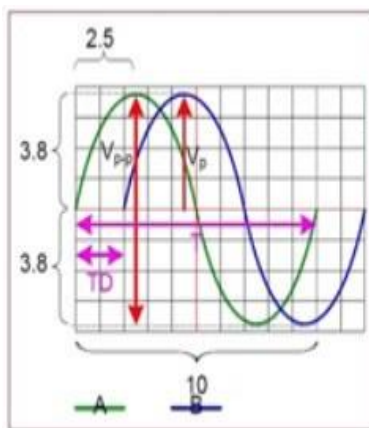
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Hz

Amplitude (V_{pp}) = no of vertical divisions x voltage per division at knob in volts

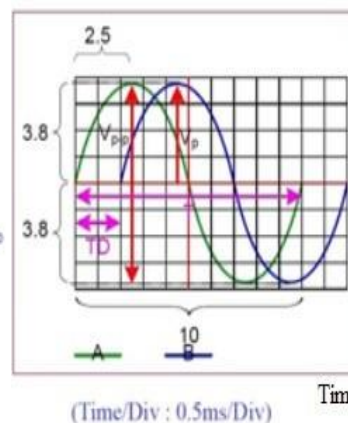
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volts



1 cycle = 10 div
 $TD = 2 \text{ div}$
 Therefore,
 1 cycle : 10 div = 360°
 1 div = $360^\circ / 10 = 36^\circ$
 2 div = $2 \times 36^\circ = 72^\circ$

Phase Difference



a) Period, T
 $T = (\text{Time/Div}) \times (\text{no. div/cycle})$
 $= 0.5\text{ms/div} \times 10$
 $= 5\text{ms}$

b) Frequency, f
 $f = 1/T$
 $= 1/5\text{ms}$
 $= 200 \text{ Hz}$

Time period and frequency

1 Cycle = ----- div

Difference = ----- div

1 cycle = no of div ----- = 360° Therefore

1 div = $360^\circ / \text{no of div}$ ----- =

-----div = -----x ----- =

RESULT: For the given AC voltage waveform Amplitude, Peak-to-Peak voltage, Frequency and Time Period are measured successfully.

Signature of staff

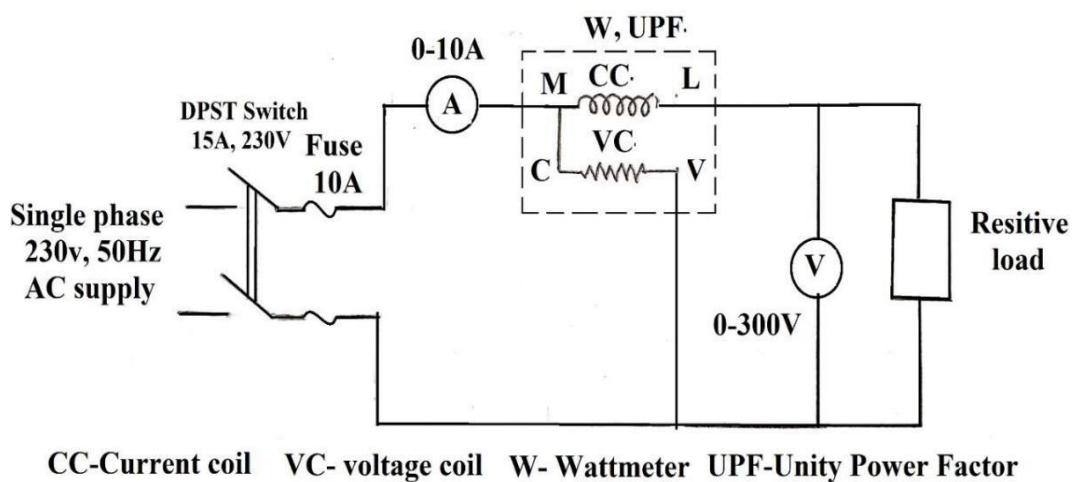
1. Measure the voltage, current, power using relevant measuring instruments in a Single phase load. Date:

Aim: Measure the Current, voltage and power using relevant measuring instruments in single phase load.

Apparatus required:

Sl No.	Particulars	Range	Quantity
1	Voltmeter	0-300V, MI(AC)	1
2	Ammeter	0-10A, MI (AC)	1
3	Wattmeter(UPF)	10 A, 300V UPF	1
4	DPST switch	-	1
5	Resistive load	single phase	1
6	Connecting wires	-	As required

Circuit diagram



Wattmeter constant $K = \frac{V(\text{selected}) \times I(\text{selected}) \times \cos\phi}{\text{Full Scale reading of wattmeter}} =$

Full Scale reading of wattmeter

Procedure

1. Connections are made as per the circuit diagram.
2. Switch on the single phase power supply and then close the DPST switch.
3. Apply the load by Turning ON the switches of resistive load and note down the readings of Ammeter, Voltmeter and Wattmeter.

4. Repeat the step 3 by increasing the load and note down the readings.
5. Open the DPST switch and turn OFF the supply.

Tabular column:

<u>Sl.No.</u>	Current in Amps	Voltage V in volts	Power P in watts $\times K$

Result: Measurement of Current, voltage and power is measured using relevant measuring instruments in Single Phase load.

Signature of staff

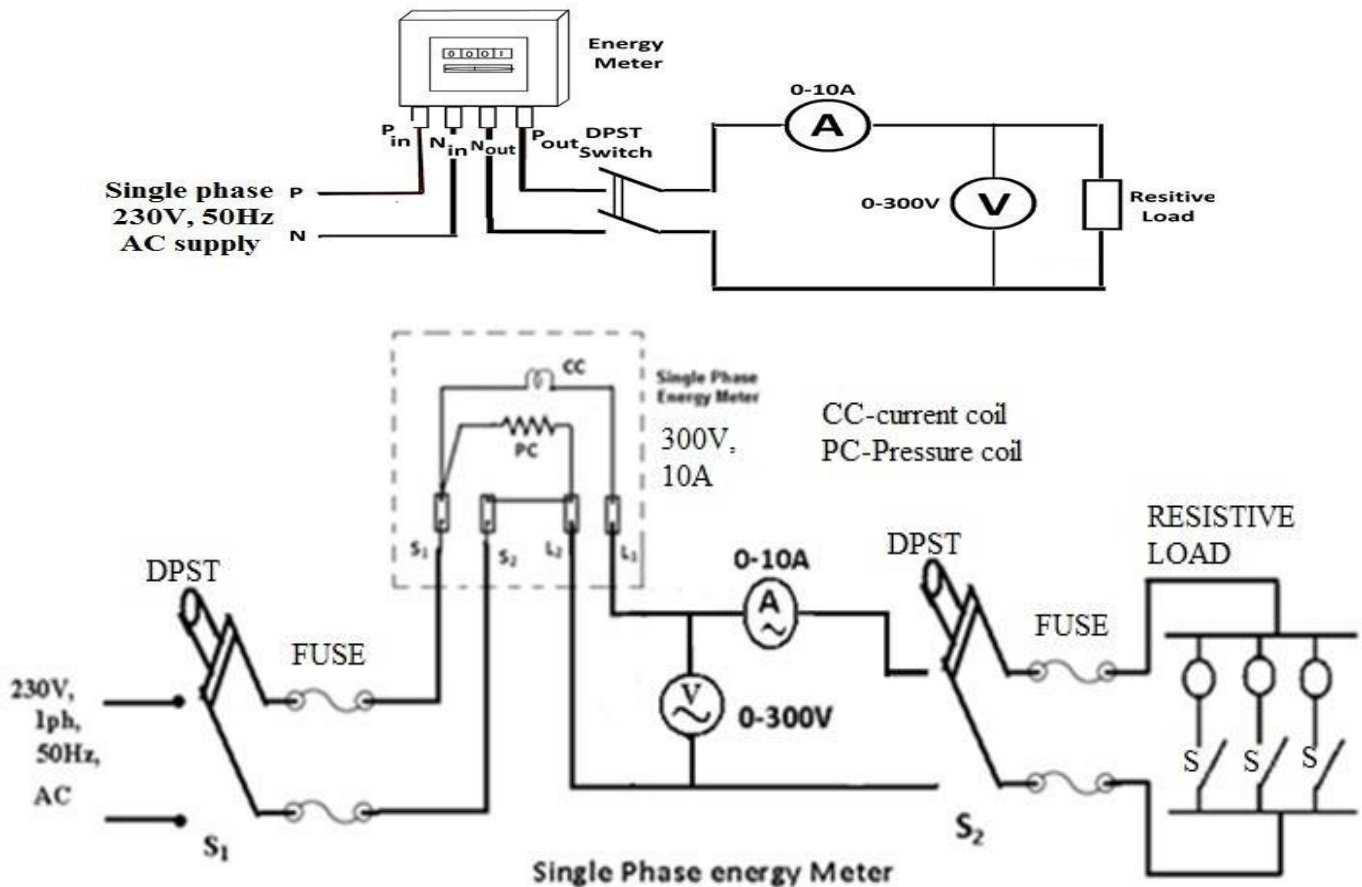
Measure single phase energy using relevant measuring instruments in a Single-phase load. Date:

Aim: Measure single phase energy using relevant measuring instruments in a single phase load.

Apparatus required

<u>Sl no.</u>	Particulars	Range	Quantity
1	Voltmeter	0-300V	1
2	Ammeter	0-10A	1
3	Energy meter	-	1
4	DPST switch	-	1
5	Resistive load	-	1
6	Connecting wires	-	As required
7	Stop watch	-	1

Circuit diagram



Procedure:

1. Connections are made as per the circuit diagram.
2. Note down the meter reading.
3. Switch on the single phase power supply and then close the DPST switch.
4. Apply the load by Turning ON the switches of resistive load and at the same time start the stop watch and note down the readings of Ammeter, Voltmeter.
5. Turn OFF the load by opening the DPST switch and at the same time stop the stopwatch and note down the reading of stopwatch.
6. Note down the final meter reading and Switch OFF the supply.



Sl No.	Current I in Amps	Voltage V in Volts	Time T in Seconds	Previous meter reading PR	Final meter reading FR	Energy recorded E $E_r = FR - PR$ in kWh
1						
2						
3						

Calculation

Energy consumed (theoretically) $E_{th} = (V \times I \times T) / (3600 \times 1000)$ in kWh

Result: Single phase energy is measured by energy meter and also calculated by using formula.

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Measure the voltages in three phase supply.

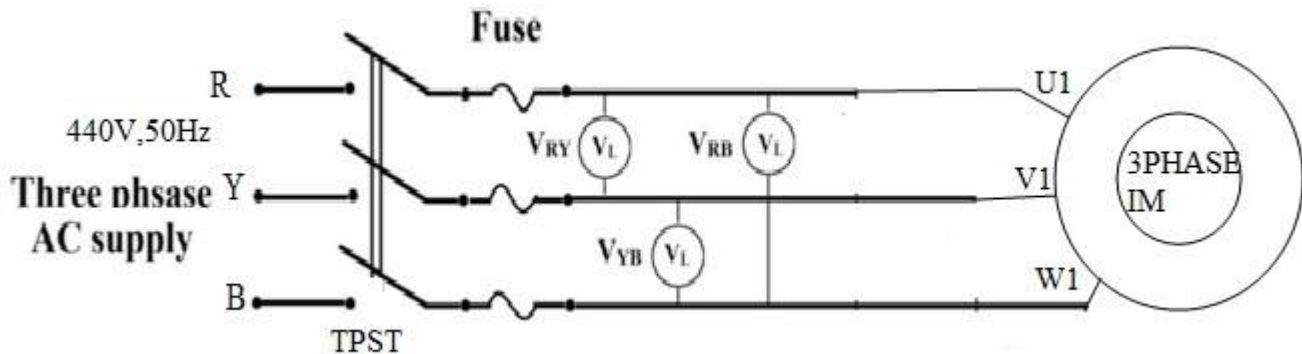
Date:

Aim: To measure the three phase voltages by using the voltmeter.

Apparatus required:

1. phase four wire power supply
2. 0-600V, Voltmeters, 3 Nos
3. Connecting wires

Circuit diagram:



Procedure

1. Connections are made as per the diagram
2. Turn on the three phase supply.
3. Close the TPST switch.
4. Measure the phase voltages and line voltages by connecting the voltmeters as shown in the circuit diagram.
5. Open TPST switch.
6. Turn OFF the Three phase power supply.

Tabular column:

Line Voltages(V_L)			Phase Voltages(V_P)		
V_{RY}	V_{YB}	V_{RB}	V_{RN}	V_{YN}	V_{BN}

Result: The phase voltages and line voltages are measured by using voltmeter from the three phase power supply.

Signature of staff

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UNIT-3**Date:**

1. Wire up and test PVC Conduit wiring to control one lamp from two different places using suitable protective devices.

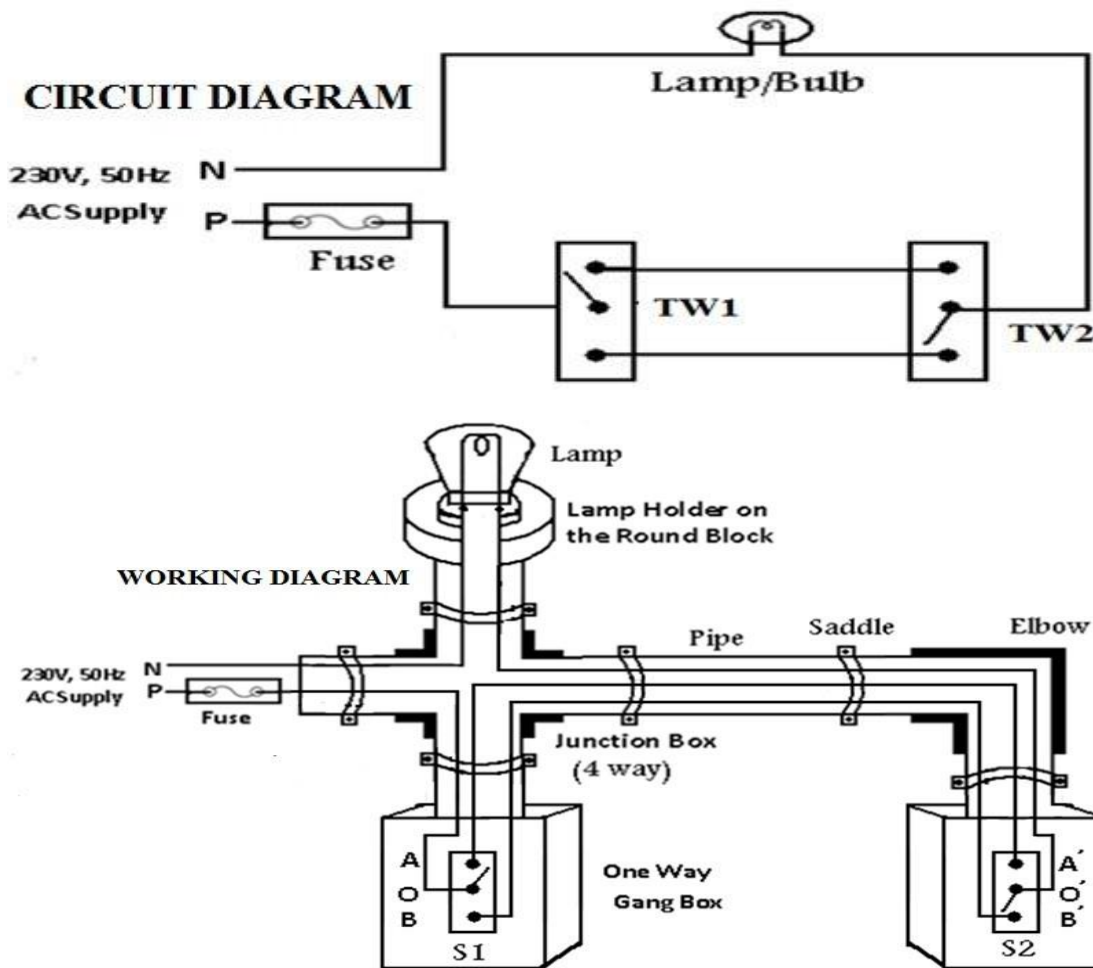
Aim: To Wire up and test PVC conduit wiring to control Lamp from two different places (staircase wiring).

Apparatus and Tools required:

<u>S</u> No.	Name of the apparatus	Specifications	Quantity
1	Wooden wiring board	2½"X2'X3"	1
2	PVC Junction box	2 way & 4 way	1 each
3	Two way switch with Gang box	5A,230V	2
4	Lamp holder	5A,230V	1
5	PVC Pipe	½"	5
6	Lamp	40W, 230V	1
7	Copper Wire	1.5SQMM	2m
8	<u>Saddels</u>	½"	6
9	Wooden Screws	¾"	20
10	Screws for Gang box	2"	12
11	Tool Kit	-	1set

Operations to be carried out

1. Marking
2. Cutting
3. Clamping
4. Skinning
5. Wiring
6. Checking
7. Testing
8. Finishing

WIRING DIAGRAM:**PROCEDURE**

1. The wiring diagram is roughly drawn on the wooden board.
2. Connections are made as shown in the circuit diagram.
3. The two way switches are connected in series with the phase only.
4. The switches and lamps are fixed neatly and tightly.
5. Check the connection once again by the test lamp for proper connections.
6. Switch on the supply and close the DPST switch.
7. Switch on the switches as per the tabular column and verify the results.
8. Switch off all switches, open DPST switch and turnoff the power supply.

TABULAR COLUMN

Switch Positions		Lamp Condition
TW1	TW2	
OA	O'A'	ON
OB	O'B'	ON
OA	O'B'	OFF
OB	O'A'	OFF

RESULT: PVC conduit wiring is done as per the wiring diagram to control the Lamp from two Different places and tested according tabular column.

Signature of staff

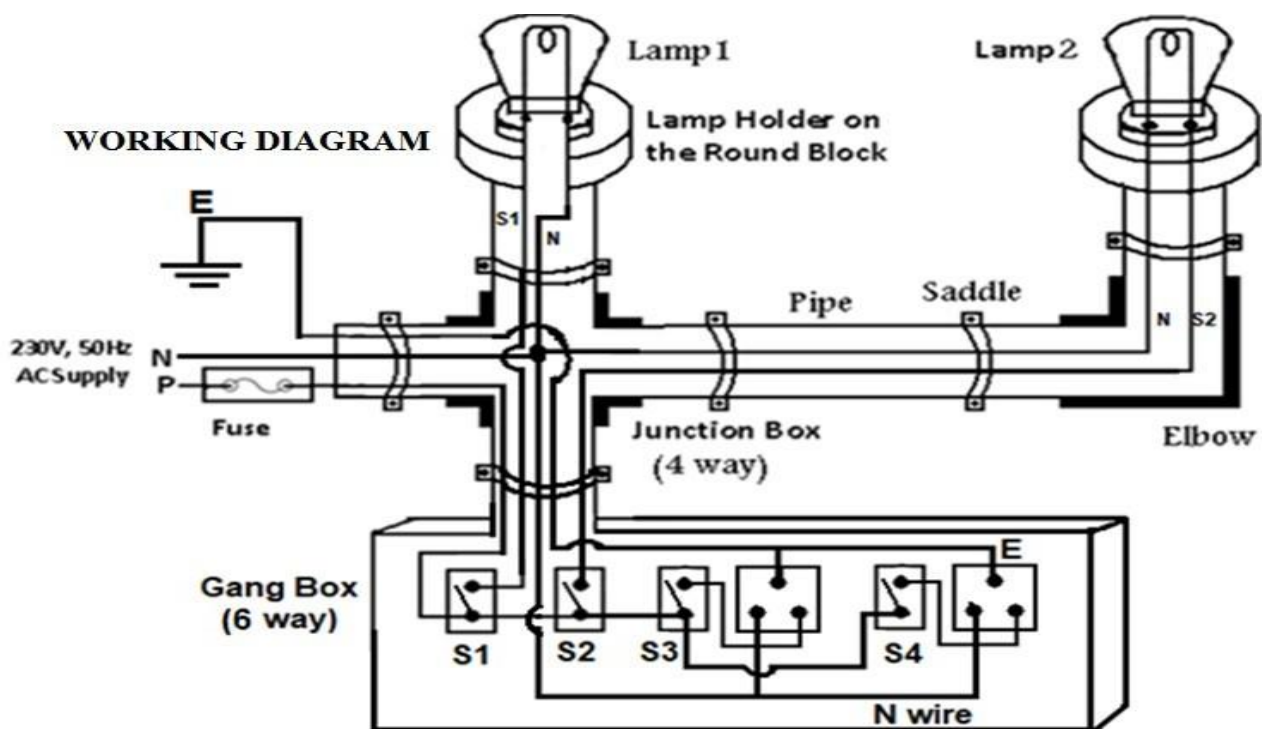
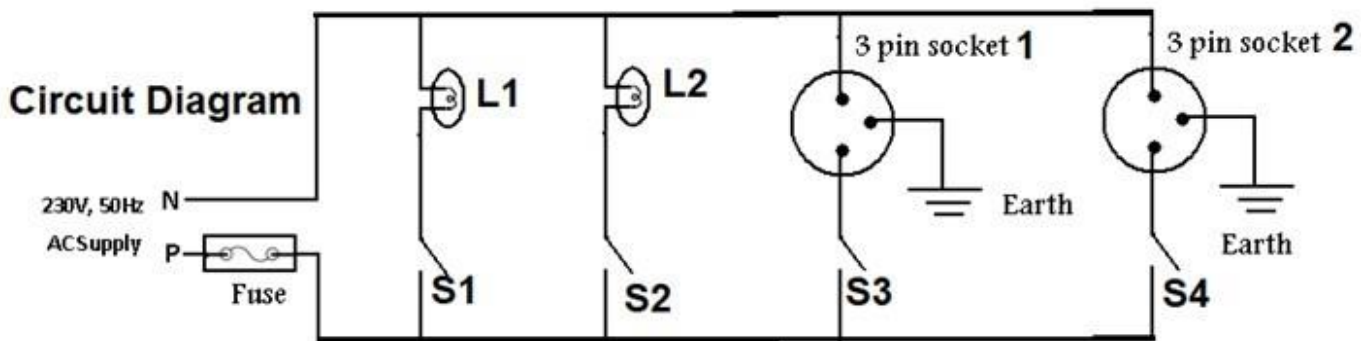
Wire up and test PVC Conduit wiring to control of 2 sockets and 2 lamps.

Date;

Aim: Wire up and test PVC conduit wiring to control of two lamps and two sockets independently.

Apparatus and Tools required:

SL No.	Name of the apparatus	Quantity	SL no	Name of the tools	Quantity
1	DPST Switch	01	1	Screw driver	01
2	Incandescent Lamp 60W	02	2	Cutting <u>plier</u>	01
3	Lamp holder	02	3	Wire stripper	01
4	Teak wood round block	04	4	Tester	01
5	Teak wood board	01			
6	Fuse cutout	02			
7	Half inch PVC pipe	12			
8	3-pin socket	02			
9	Wooden screw	24			
10	Junction box(4 way, 3 way)	03,01			
11	Test lamp	01			

CIRCUIT DIAGRAM:**PROCEDURE:**

- 1). Draw the layout of the ckt on the wiring board.
- 2). Fix the conduit pipes, elbow, junction box, etc. on the board using saddles and screws.
- 3). Draw the wires straightly by screwing through pipes.
- 4). Connect and fix the Gang Box, SP switches, Round Blocks, Lamp Holders, Sockets, Fuse, etc.
- 5). After completing the wiring, fix the lamps on to the lamp holders.
- 6). Give supply.
- 7). Observe and verify the results for various switch positions.
- 8). Also verify the connectivity of the earth connection using a lamp.

TABULAR COLUMN:

Sl no.	Switch 1	Switch 2	Switch 3	Switch 4	Lamp 1	Lamp 2	Socket 1	Socket 2
1	ON	ON	OFF	OFF				
2	OFF	OFF	ON	ON				
3	ON	ON	ON	ON				
4	OFF	OFF	OFF	OFF				

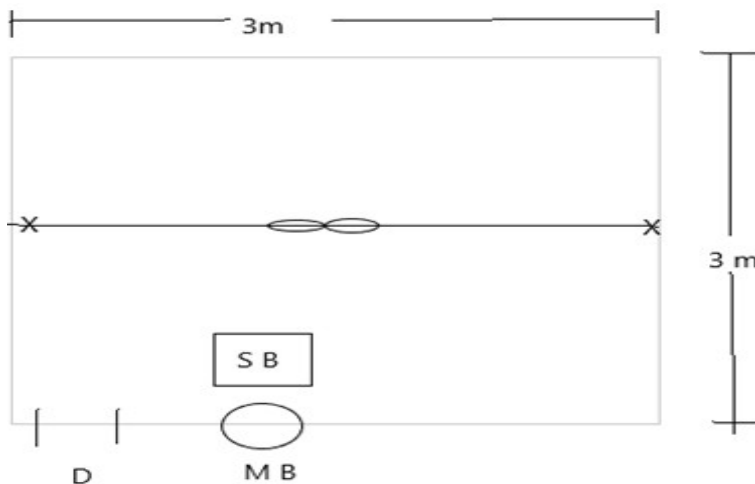
RESULT: PVC conduit wiring is done as per the wiring diagram to control the two Lamps and Two Sockets independently and tested according to the tabular column.

Signature of staff

Prepare the estimation and plan

Date:

Aim: Plan and estimate the cost of electrical wiring for one 3mx3m room consisting of 2 CFL 1ceiling fan, 2 three pin sockets.

**LOAD CALCULATION**

1. Light load = $25\text{w} \times 2 = 50\text{w}$
2. Fan load = $60\text{w} \times 1 = 60\text{w}$
- 3.5A, socket = $100\text{w} \times 2 = 200\text{w}$

Total connected load = 310w

Load current = $P/V = 310 / 230 = 1.35\text{A}$

Size of cable

1sq mm, single core copper conductor, PVC insulated cable

Size of PVC pipe: ¾" dia, 2mm thick PVC Pipe.

Length of pipe:

Horizontal run= $0.5 + 1.5 + 3 = 5\text{m}$

Vertical run= $0.5 + 1 + 1.5 + .5 + 0.5 = 4\text{m}$

10% extra = 0.9% $9 + 0.9 = 9.9\text{m} = 10\text{m}$

Length of cable = 3x length of pipe = $3 \times 10 = 30\text{m}$

List of materials

1. 1.5sq mm single core cu conductor=30m

2. ¾" dia, 2mm thick PVC pipe=10m

3. PVC bends for ¾" dia pipe=6No

4. PVC Junction box =3No

5. DP main switch, 16A, 250V=1 No

6. Single phase MCB, 3A, 250V=1No

7. SP Switches, 6A, 250V=5No

8. Wall plug sockets 6A, 250V=2No

9. Porcelain fuse cutout 15A, 250V=1No

10. Ceiling rose 6A 250V=1 No

11. Nylon round block =3No

12. Angle holder 6A 250V=2No

13. Teak Wood board of size 12"x18"=1No

14. Teak Wood board of size 6"x10"=1No

15. Plym sheet of size 6"x10"=1No

16. GI Pipe 40mm dia, 3mm thick=1.5m

17. 8 SWG GI Wire=1kg

Detailed estimation

Sl No	Material With specification.	Unit	Quantity	Rate	Cost
1	1.5sq mm single core cu conductor	m	30		
2	¾" dia , 2mm thick PVC pipe	m	10		
3	PVC bends for ¾" dia pipe	No	6		
4	PVC Junction box	No	3		
5	DP main switch , 16A,250V	No	1		
6	Single phase MCB , 3A,250V	No	1		
7	SP Switches, 6A,250V	No	5		
8	Wall plug sockets 6A , 250V	No	2		
9	Porcelain fuse cutout 15A,250V	No	1		
10	Ceiling rose 6A250V	No	1		
11	Nylon round block	No	3		
12	Angle holder 6A 250V	No	2		
13	Teak Wood board of size 12"x18"	No	1		
14	Teak Wood board of size 6"x10"	No	1		
15	Hylem sheet of size 6"x10"	No	1		
16	GI Pipe 40mm dia , 3mm thick	m	1.5		
17	8 SWG GI Wire	kg	1		

RESULT: the plan and estimation is prepared

Signature of staff

UNIT-4**Date:**

1. Connect the Single- phase transformer as Step-Up, Step-Down transformer and verify the transformation ratio.

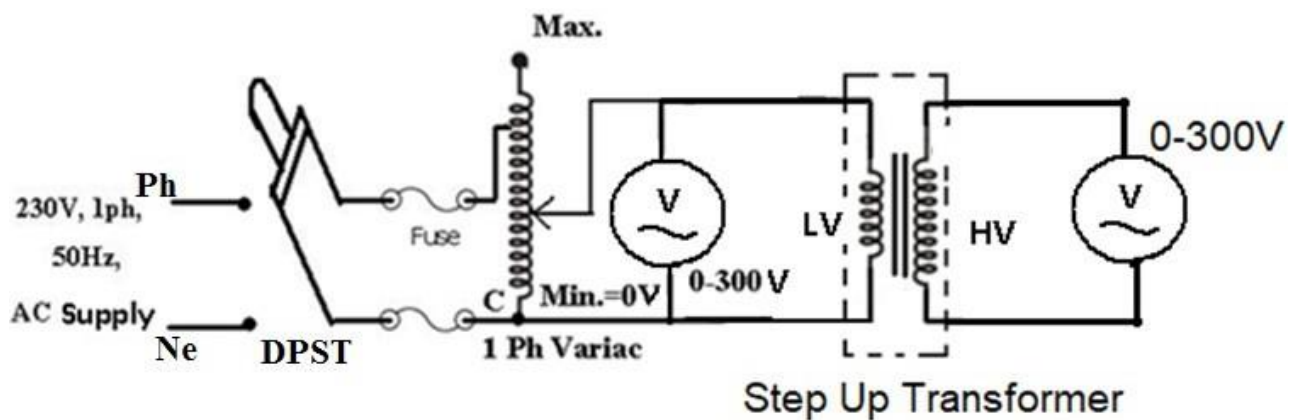
Aim: To connect transformers as step up and step down transformers and verify Transformation ratio.

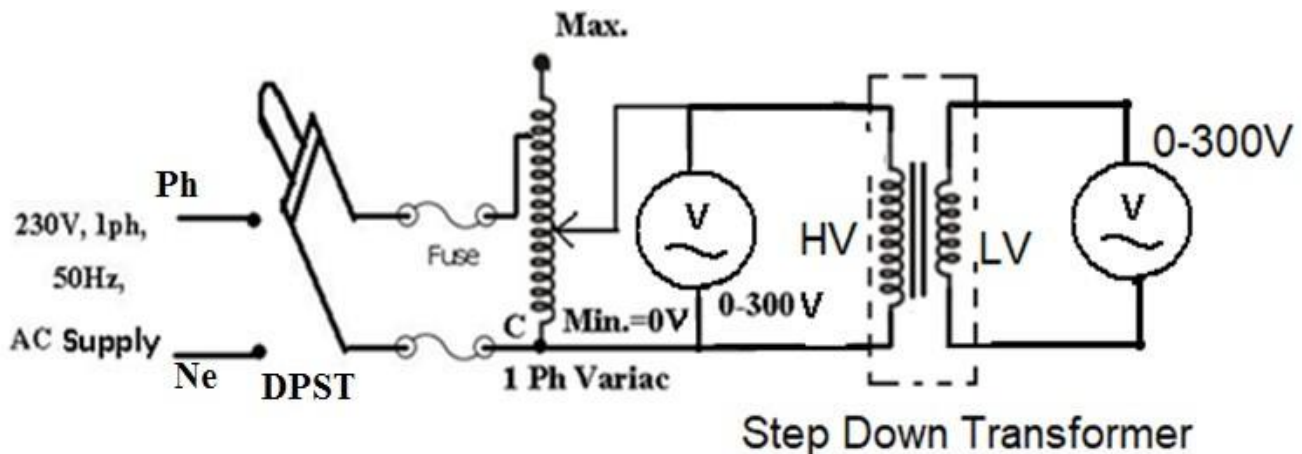
Apparatus

SLNo.	Item with Specification	Quantity
1	Transformer with different tapings on primary and secondary	01
2	Voltmeter 0-300 V AC(MI)	01
3	Auto transformer single phase	01
4	Connecting wires	required

Theory:

Transformer is a device which works on the principle of mutual induction and transfers electrical power from one circuit to another circuit without change in the frequency. It consists of two windings. Primary winding and secondary winding. There will not be electrical connection between primary and secondary.

Circuit Diagram



Procedure:

Step-Up Transformer

1. Connections are made as per the circuit diagram.
2. It is made sure that auto transformer is Zero position.
3. The power supply is switched on and the wiper of the auto transformer is varied and the voltage is set to 110V.
4. The voltage across the primary and secondary of the transformer is noted down.
5. The wiper of the autotransformer is brought to zero position and the supply is switched off.

Step-Down Transformer

1. Connections are made as per the circuit diagram.
2. It is made sure that auto transformer is Zero position.
3. The power supply is switched on and the wiper of the auto transformer is varied and the voltage is set to 220V.
4. The voltage across the primary and secondary of the transformer is noted down.
5. The wiper of the autotransformer is brought to zero position and the supply is switched off.

Tabular Column

For step up configuration:-

Sl No	V_1 volt	V_2 volt	V_2/V_1	Marked value of V_2/V_1

Average value of Transformation ratio = $V_2/V_1 =$

For step down configuration:-

Sl No	V_1 volt	V_2 volt	V_2/V_1	Marked value of V_2/V_1

Average value of Transformation ratio = V_2/V_1 =

V_2 = Voltage across the secondary winding.

V_1 = Voltage across the primary winding,

N_2 = number of turns o the primary winding,

N_1 = number of turns on the secondary winding.

Result: The transformation ratio of single phase transformer is calculated for both step-up and step-down transformer and it is found that theoretical value and practical value are approximately equal.

Signature of staff

1. Construct a suitable circuit to start and reverse the direction of three phase induction motor using DOL/ Star-delta starter.

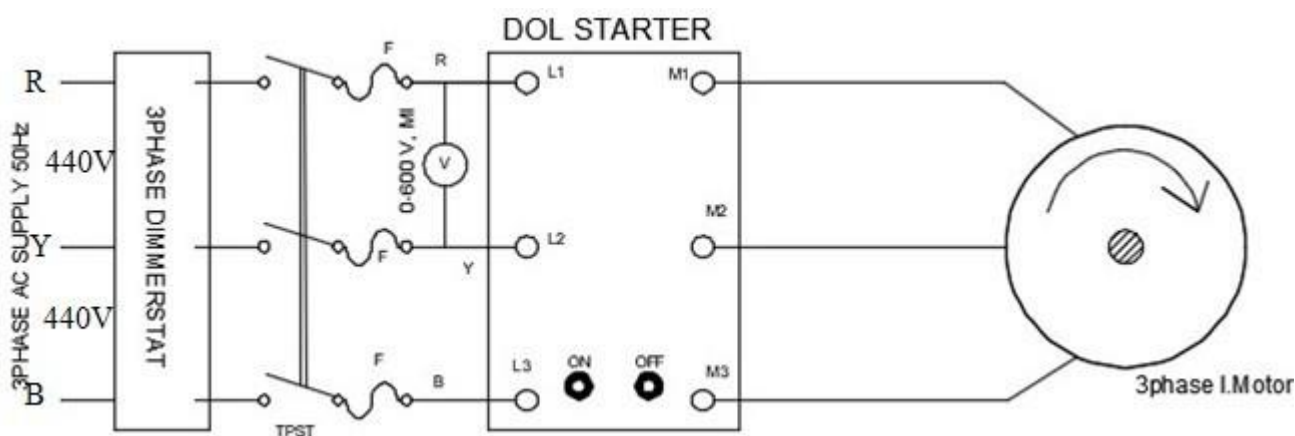
Date:

Aim: To connect and run 3-Phase Induction Motor in forward and reverse direction using DOL starter.

Apparatus required

SLNo.	Item with Specification	Quantity
1	3 Phase 440V, 5 HP, Induction motor	1
2	DOL Starter	1
3	Screwdriver	1
4	Wire stripper	1
5	Connecting wires	as required

Circuit Diagram



Theory:

Small motors are started using Direct on Line (DOL) Starter Full supply voltage is applied to the motor during starting. Thermal over load protection is normally provided. Motors having a rating up to 5 HP can be started using this type of starter. For Motors having rating of 5 to 20 HP star Delta Starter is used. Initially the stator windings of the motor are connected in star and reduced voltage is applied. After some time the windings will be connected in Delta

Procedure:

1. Connect the circuit elements as shown in the circuit diagram.
2. Close the Supply switch.
3. Switch on the start button (Green) of the starter.
4. Observe the direction of rotation of the motor.
5. Switch off the starter (Red).
6. Open the supply switch
7. Interchange the connections of any two terminals of the supply.
8. Repeat the steps 2 to 5

Precautions:

1. Before connecting check and ensure that motor and the starter are in good condition.
2. The terminals and screws are tightened firmly.
3. The starter should not be operated with its cover in the open condition
4. Before switching on the mains it is ensured that nobody is touching the starter or the motor.
5. While reversing the direction of rotation of Motor, allow the motor to come to standstill position before attempting to operate in the opposite direction. This avoids the strain on the motor.

Tabular Column

SL NO	PHASE SEQUENCE	DIRECTION OF ROTATION
1	RYB	
2	RBY	

Result: It is observed that the direction of rotation of motor can be reversed by changing the phase sequence.

Signature of staff

Testing Condition of charging and discharging of a Lead-acid battery

Date;

Step 1: Test the battery voltage. Set the voltmeter dial to DC (direct current) volts. Place the voltmeter leads on the positive and negative battery posts. Read the voltage. The voltage should read greater than 12.6 volts and less than 14.0volts.

Step 2 : Charge the battery. Perform a high-rate charge test. Hook positive and negative battery cables onto the battery posts and set the charger to fast charge or boost. Take a voltage reading with the voltmeter. If the voltage rate quickly climbs and exceeds 14.0 volts, reading as high as volts, the battery cannot contain the voltage and is defective. Replace the battery.

Step 3 : Perform a slow charge test. Slow charge the battery for several hours. Periodically inspect the battery. If it is bubbling, spewing and has spattered battery acid on the cover, the battery is defective and must be replaced.

Perform a load test. Slow charge the battery for several hours, overnight if possible. If the battery has passed all the other tests, is clean and dry and the voltage reads correctly, attach a load tester and read the scale. The needle should be in the green zone. Turn the load test knob. This allows current to flow into a carbon pile where it is turned to heat. Do this for an approximate time it would take to start a vehicle. The needle should stay in the green zone. Repeat the test two or three more times. Each time the needle should remain in the green zone. If it drops into the red, replace the battery.

signature of staff

Sizing of UPS**Date:**

Estimate the UPS rating for a computer lab with 50 computers/ domestic.

Aim: To Estimate the UPS rating for a computer lab with 50 computers.

Procedure:

1. List all equipment to be protected by the UPS
2. Check the wattage of each desktop computer and convert the value to the VA rating.
3. Multiply the VA by the number of equipment to get the VA Total.
4. Multiply the total by safety factor 1.2 to get the grand total to the VA total.
5. Use the grand total to select a UPS.
6. Assume power factor as 0.8.
7. Assume all the computers are of same wattage.

An uninterruptible power supply or uninterruptible power source (UPS) is an electrical apparatus that provides emergency power to a load when the input power source or mains power fails.

Types of UPS:

1. Offline/Standby UPS
2. Online/Double Conversion UPS
3. Line-Interactive PS

Applications:

1. Healthcare: hospitals, clinics and homes. Telecommunications.
2. Special projects (events)
3. Data centers
4. Industries
5. Telecommunications
6. Hospitals
7. Banks and insurance

Tabular column

Equipment	wattage	VA conversion	Qty	VA Total
Desktop computer	200	$200/0.8(\text{power factor})=250$	50	12500
GRAND TOTAL				15000

1.2 times of total KVA $12500 \times 1.2 = 15000 \text{ VA}$

$\text{KVA} = \text{VA}/1000 \quad 15000/1000 = 15 \text{ KVA}$

Estimation of 15kva inverters available in the market.

Sl no	Company name	Capacity	Units	price
01	Microteki-MAXX	15KVA	01	₹ 1,90,500.00
02	Olympus	15KVA	01	₹ 2,46,580.00
03	APC	15KVA	01	₹ 2,75,000.00

Result: UPS requirement for computer lab is estimated with required quantity.

Signature of staff

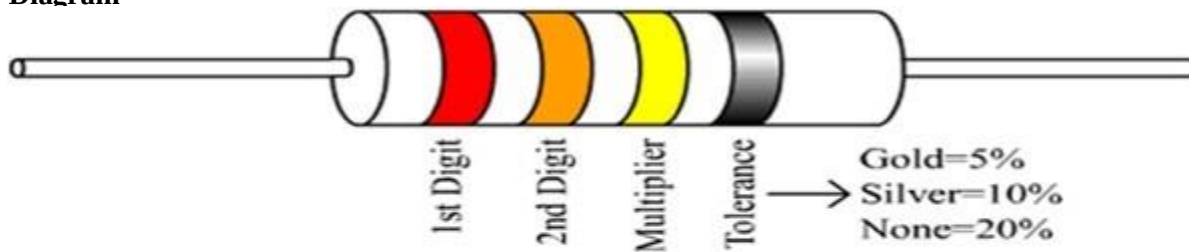
UNIT-5**Date:**

Determine the value of resistance by color code and compare it with multimeter readings.

Aim: To determine the value of resistance by color coding method and compare it with multimeter readings.

Apparatus required

SLNo.	Item with Specification	Quantity
	Digital <u>multimeter</u>	1
2	resistors	Few
3	Patch cords	1 Set

Diagram

	Color	Digit	MULTIPLIER	TOLERANCE
	Black	0	$10^0 = 1$	
	Brown	1	$10^1 = 10$	
	Red	2	$10^2 = 100$	
	Orange	3	$10^3 = 1000$	
	Yellow	4	$10^4 = 10000$	
	Green	5	$10^5 = 100000$	
	Blue	6	$10^6 = 1000000$	
	Violet	7	$10^7 = 10000000$	
	Grey	8	$10^8 = 100000000$	
	White	9	$10^9 = 1000000000$	
	Gold		$10^{-1} = 0.1$	5%
	Silver		$10^{-2} = 0.01$	10%
	(none)			20%

Calculate resistor value using color code:

The first two bands always denote the first two digits of the resistance value in ohms. On a three or four-band resistor, the third band represents the multiplier.

This multiplier will basically shift your decimal place around to change your value from mega ohms to milliohms and anywhere in between. The fourth color band signifies tolerance. Keep in mind that if this band is absent and you are looking at a three-band resistor, the default tolerance is $\pm 20\%$.

Procedure

1. Given resistors of different color bands are taken.
2. Using the color code table, the values of resistor is calculated.
3. The tolerance value of higher range and lower range is found out.
4. Digital multimeter is taken and the value of resistor is measured by setting the selector switch to Resistance range.
5. The difference in measured value and calculated value is noted.

Tabular column

BAND	COLOR CODE	NUMERICAL VALUE	MULTIMETER READING
1 ST BAND			
2 ND BAND			
3 RD BAND			
4 TH BAND			

The Resister value is-----

BAND	COLOR CODE	NUMERICAL VALUE	MULTIMETER READING
1 ST BAND			
2 ND BAND			
3 RD BAND			
4 TH BAND			

The Resister value is-----

BAND	COLOR CODE	NUMERICAL VALUE	MULTIMETER READING
1 ST BAND			
2 ND BAND			
3 RD BAND			
4 TH BAND			

The Resister value is-----

Result:

The given resistor value is calculated using color coding and verified using digital multimeter.

Signature of staff

Identify the terminals of a Diode and test the diode for its condition.

Date:

Aim: Identify the terminals of a diode and test the diode for its condition.

Apparatus Required

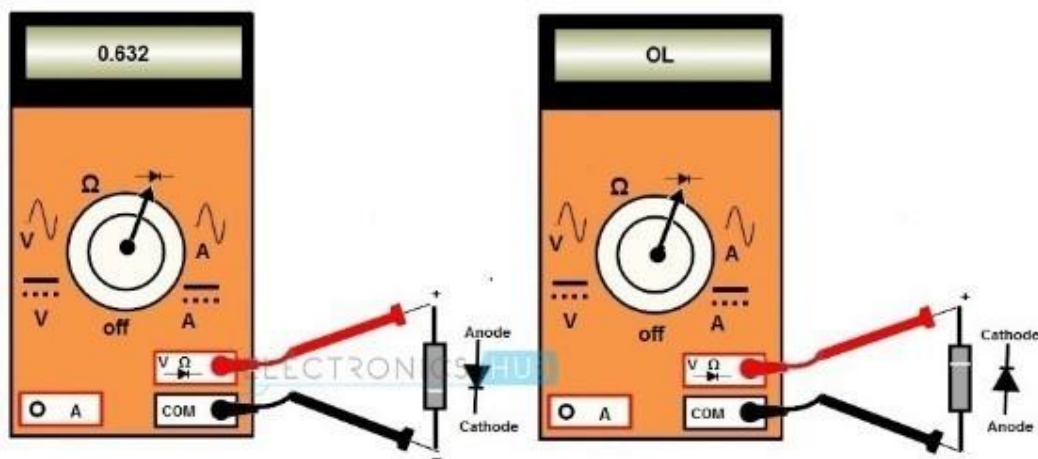
Sl No	Name	Type	Range/Specification	Qty
1	Digital multimeter			1
2	Test probes			1
3	Diode	1N4007	1 A, 1000 V	1

Terminal Identification; Before testing of the diode we have to identify the terminals of the diode that is anode and cathode. Most of the PN diodes have the white-band on its body and this white-band side terminal is the cathode. And the remaining one is anode. Some diodes may have a different color band, but the color band side terminal is the cathode.

Procedure:

Identify the terminals anode and cathode of the diode.

1. Keep the digital multimeter (DMM) in diode checking mode by rotating the central knob to the place where the diode symbol is indicated. In this mode multimeter is capable to supply a current of 2mA approximately between the test leads.
2. Connect the red probe to the anode and black probe to the cathode. This means diode is forward-biased.
3. Observe the reading on meter display. If the displayed voltage value is in between 0.6 to 0.7 (since it is silicon diode) then the diode is healthy and perfect. For germanium diodes this value is in between 0.25 to 0.3.
4. Now reverse the terminals of the meter that means connect the red probe to cathode and black to anode. This is the reverse biased condition of the diode where no current flows through it. Hence the meter should read OL (which is equivalent to open circuit) if the diode is healthy



Testing of DIODE

Procedure:

Identify the terminals anode and cathode of the diode.

5. Keep the digital multimeter (DMM) in diode checking mode by rotating the central knob to the place where the diode symbol is indicated. In this mode multimeter is capable to supply a current of 2mA approximately between the test leads.
6. Connect the red probe to the anode and black probe to the cathode. This means diode is forward-biased.
7. Observe the reading on meter display. If the displayed voltage value is in between 0.6 to 0.7 (since it is silicon diode) then the diode is healthy and perfect. For germanium diodes this value is in between 0.25 to 0.3.
8. Now reverse the terminals of the meter that means connect the red probe to cathode and black to anode.
9. This is the reverse biased condition of the diode where no current flows through it. Hence the meter should read OL (which is equivalent to open circuit) if the diode is healthy.

Tabular column:

Sl. No	Multimeter connection with Diode	Reading.	Remarks
1	Red to Anode, Black to cathode.		
2	Red to Cathode, Black to Anode		

Result: The given Diode 1N4007 is in good condition.

Signature of staff

Construct and test bridge rectifiers using semiconductor diode and rectifier IC. Compare the waveforms using CRO.

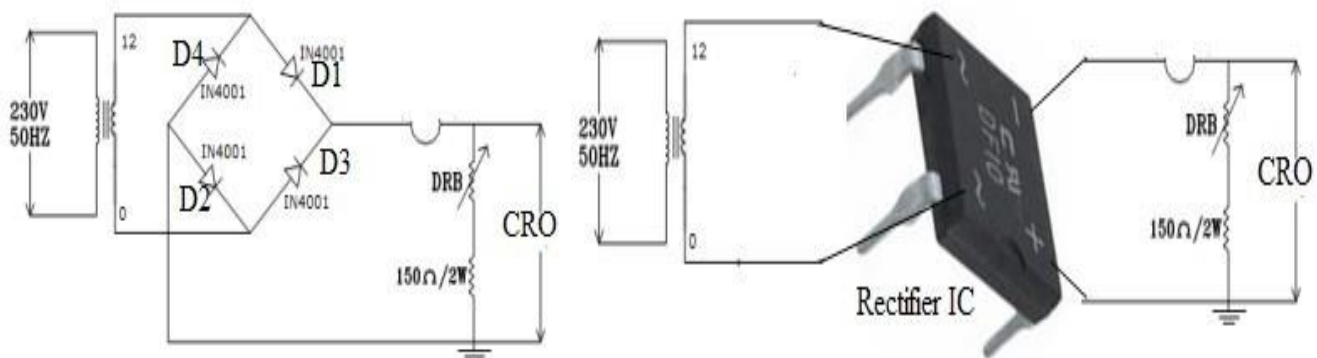
Date:

Aim: To Construct and test bridge rectifiers using semiconductor diode and rectifier IC. Compare the waveforms using CRO.

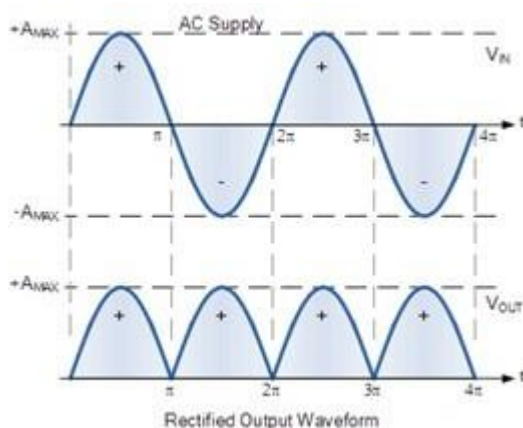
APPARATUS REQUIRED

S1 No	Name	Type	Range/Specification	Qty
1	Transformer		12-0-12, 3 amp	01
2	CRO		25 M Hz	01
3	BNC			02
4	Diode	1N4007		04
5	Resistor	Carbon	1 K Ω , $\frac{1}{2}$ watt	01
6	IC	RB 156	1.5A, 560V	01
7	Connecting wires			Lump sum

Circuit Diagram



Waveform



SL.NO	With Diode		With IC	
	VDC	VAC	VDC	VAC
1				

Procedure:

1. Connections are made as per the circuit diagram.
2. Switch on CRO and make settings.
3. Switch on power supply to the rectifier circuit and observe the wave form and trace it.
4. Take readings.
5. Switch off power supply.
6. Calculate ripple factor.

Conclusion: Waveform shows semiconductor Diode and Rectifier IC can be used as a rectifier

Signature of staff

1. Identification of transistor terminals and test.

Date:

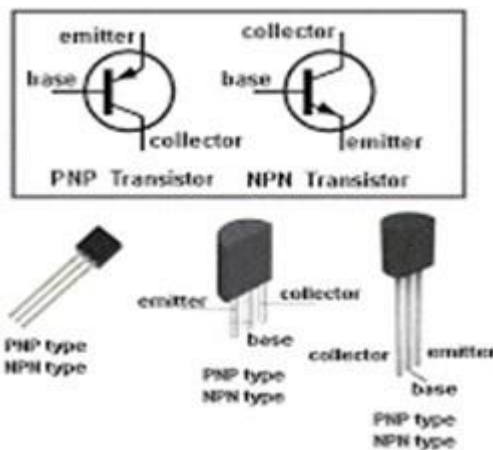
2. Construct and test the transistor as an electronic switch.

Aim: 1.To identifies and test NPN or PNP transistor and its terminal identification.

Apparatus Required:

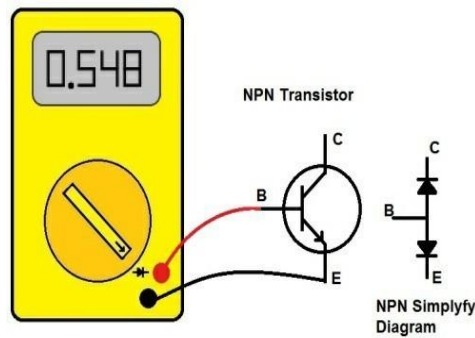
Sl No	Name	Type	Range/Specification	Qty
1	Transistor	BC547/BC107	45 V, 100 mA	01
2	Digital Multimeter			01
3	Resistors	Carbon	4.7K, 1K	01
4	DC Power supply	RPS, linear	0-30V, 2A	01
5	LED	CQY24		01
6	Bread board			01

Transistor



Steps to identify the NPN type transistor:

1. Keep the Multimeter in the Diode mode.
2. Keep the positive probe to the center pin of the transistor.
3. Touch the negative probe to the pin-1. You will see some voltage in the multimeter.
4. Similarly touch the negative probe to the pin-3 (collector) with respect to the center pin. You will see some voltage in the multimeter.
5. It will ensure that it is a NPN transistor and the center pin must be base. The logic behind this is, in NPN transistor, Emitter (E) – N type material – Equivalent to cathode of the diode. Base (B) – P type material – Equivalent to anode of the diode. Collector(C) – N type material – Equivalent to cathode of the diode.
6. Compare voltage reading between base and terminal 1 and 2, more reading with terminal must be Emitter and other one is Collector.
7. If the multimeter positive probe is connected to anode and negative probe is connected to cathode, then it will show voltage. If the connections are interchanged it will not show any value.



Steps to identify the PNP type transistor

1. Keep the Multimeter in the Diode mode.
2. Keep the positive probe to the pin-1 (Emitter) of the transistor.
3. Touch the negative probe to the center pin (Base). You will see some voltage in the multimeter.
4. Similarly touch the negative probe to the center pin (Base) with respect to the pin-3 (collector). You will see some voltage in the multimeter. (E) It will ensure that it is a PNP transistor. The logic behind this is, in PNP transistor Emitter –P type material – Equivalent to anode of the diode. Base (B) – N type material – Equivalent to cathode of the diode. Collector (C) – P type material – Equivalent to anode of the diode.
5. If the multimeter positive probe is connected to anode and negative probe is connected to cathode, then it will show voltage. If the connections are interchanged it will not show any value.

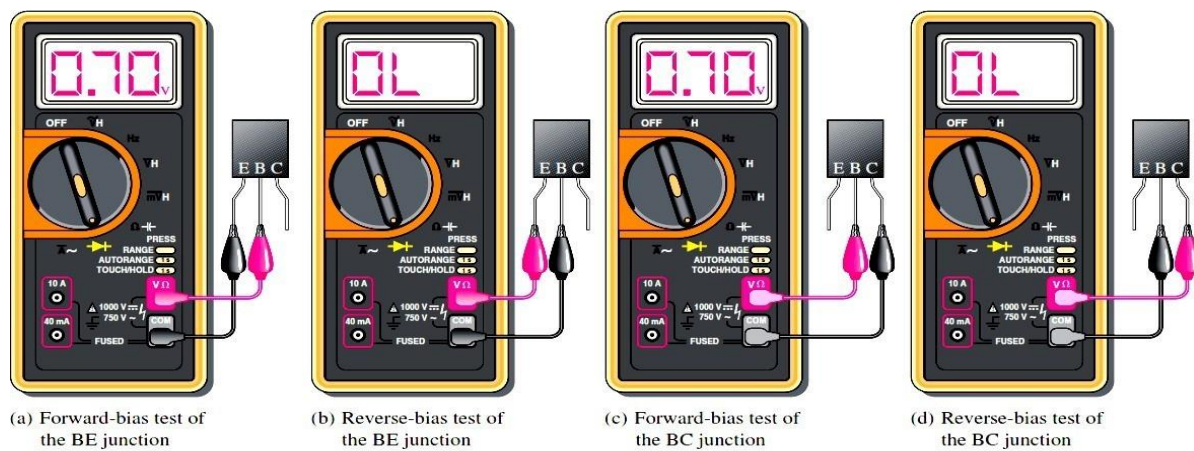


Fig : Typical DMM test of a properly functioning npn transistor. Leads are reversed for a pnp transistor.

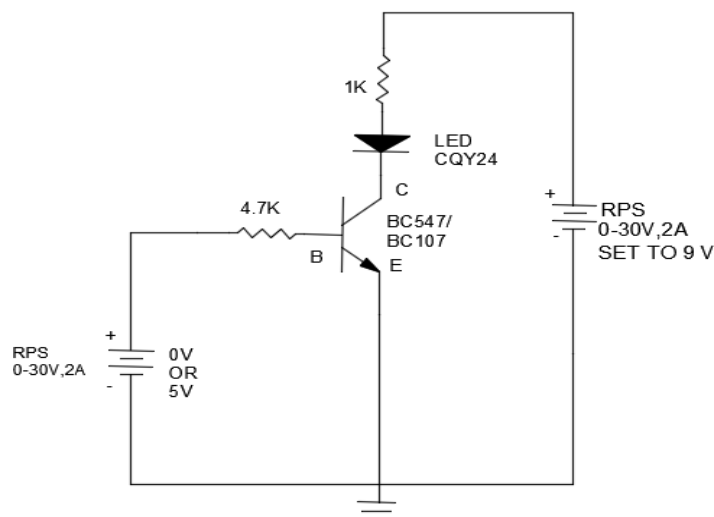
EngineeringTutorial.com

Result: The given transistor is npn type and is in good working condition.

Signature of staff

2. Construct and test the transistor as an electronic switch**Date:****Aim:** Build a circuit to test the transistor as an electronic switch.**Apparatus required**

1. RPS 0-30v -2n0s
2. BC 547/BC107 -1no
3. 4.7K resistor -1no
4. LED -1no
5. Connecting wires

Circuit Diagram**Procedure:**

1. Connections are made as per the circuit diagram.
2. Connect ZERO volts to the input terminals ie. VBE (Base to emitter voltage) and observe the Status of LED.
3. Connect FIVE to the input terminals ie. VBE (BASE to emitter voltage) and observe the status of LED.
4. Switch of the power supply.

Tabular column

Sl no	Base to emitter voltage (V_{BE}) in V	Mode of LED
1	0	OFF
2	5	ON

Result: when VBE (Base to emitter voltage) is ZERO Volts LED turned OFF and when VBE (Base to emitter voltage) FIVE volts LED is turned ON, hence transistor is Working as a switch.

Signature of staff

Test a Digital IC.**Date:**

- Identification and selection of suitable ICs for basic gates.

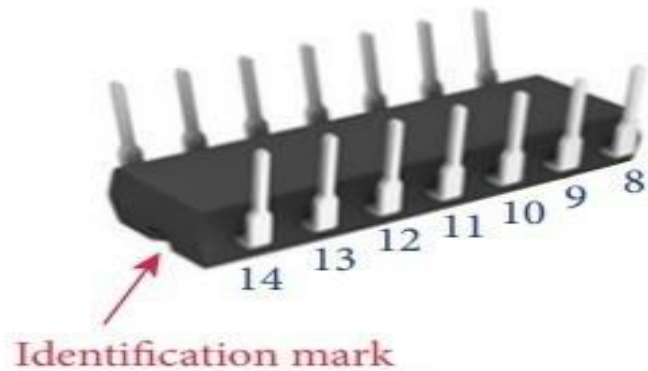
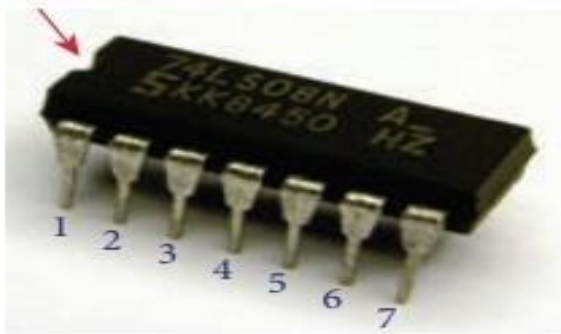
Theory: Logic gates are electronic circuits which perform logical functions on one or more inputs to produce one output. There are seven logic gates. When all the input combinations of a logic gate are written in a series and their corresponding outputs written along them, then this input/ output combination is called **Truth Table**. The following logic gates and their working are explained.

Pin Identification

Note: The chip must be inserted in the bread board in such a way that the identification mark should be on our left side. In this position, pin numbers are counted as marked in the picture above. Pin identification is the same for all chips that are mentioned below.

Pin Identification

Identification mark



Identification mark

Figure (a) Integrated circuit

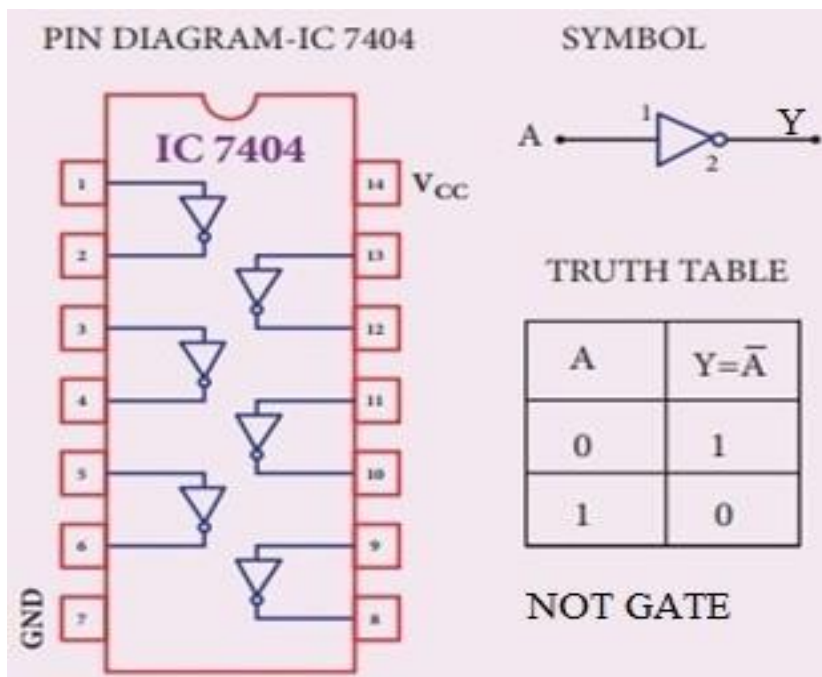
AIM: Verify NOT, AND, OR, NOR, EXOR and NAND gate operations (two inputs).

COMPONENTS REQUIRED

1. NOT gate(IC7404),
2. OR gate(IC7432),
3. NOR gate (IC 7402)
4. X-OR gate (IC 7486),
5. AND gate(IC7408),
6. NAND gate (IC 7400),
7. Digital IC trainer kit,
8. Connecting wires.

NOT Gate

NOT gate produces the complement of its input. This gate is also called an INVERTER. It always has one input and one output. Its output is 0 when input is 1 and output is 1 when input is 0.

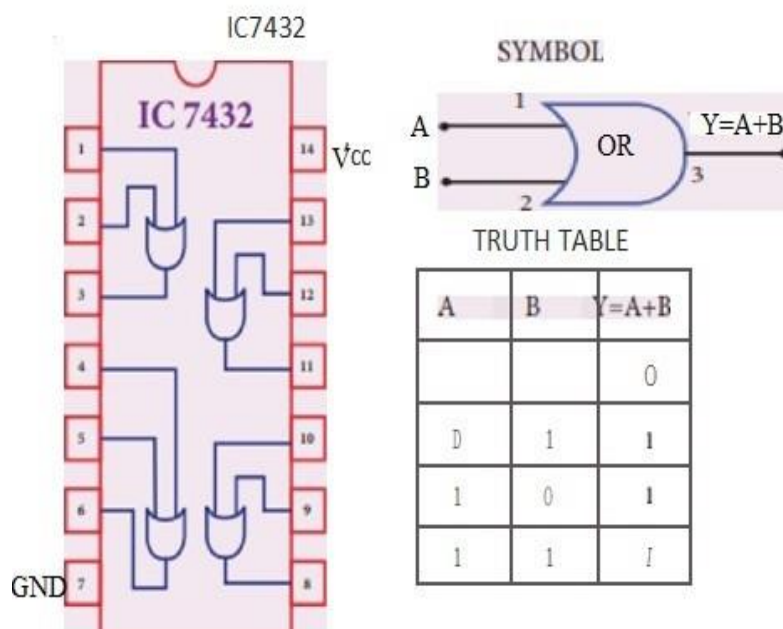


Truth Table of INVERTER Gate

INPUT A	Theoretical OUTPUT $Y = \bar{A}$	Output observed $Y_1 = \bar{A}$
0	1	
1	0	

OR Gate

OR gate produces an output as 1, when any or all its inputs are 1; otherwise the output is 0. This gate can have minimum 2 inputs but output is always one. Its output is 0 when all input are 0.

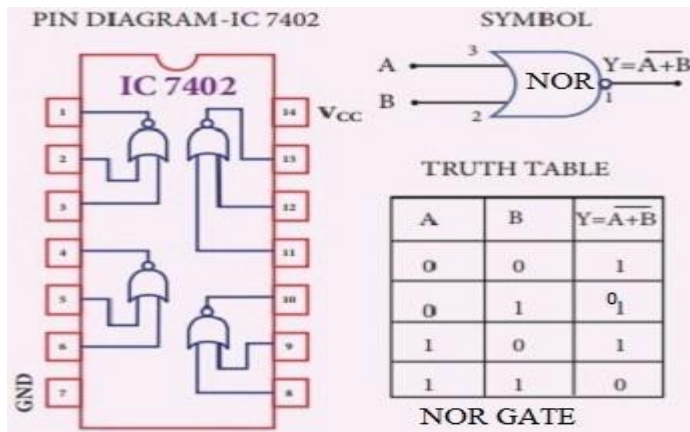


Truth Table of OR Gate

INPUTs		Theoretical OUTPUT	Output observed
A	B	$Y = A+B$	$Y_1 = A+B$
0	0	0	
0	1	1	
1	0	1	
1	1	1	

NOR Gate

NOR gate is actually a series of OR gate with NOT gate. If we connect the output of an OR gate to the input of a NOT gate, this combination will work as NOT-OR or NOR gate. Its output is 0 when any or all inputs are 1, otherwise output is 1.

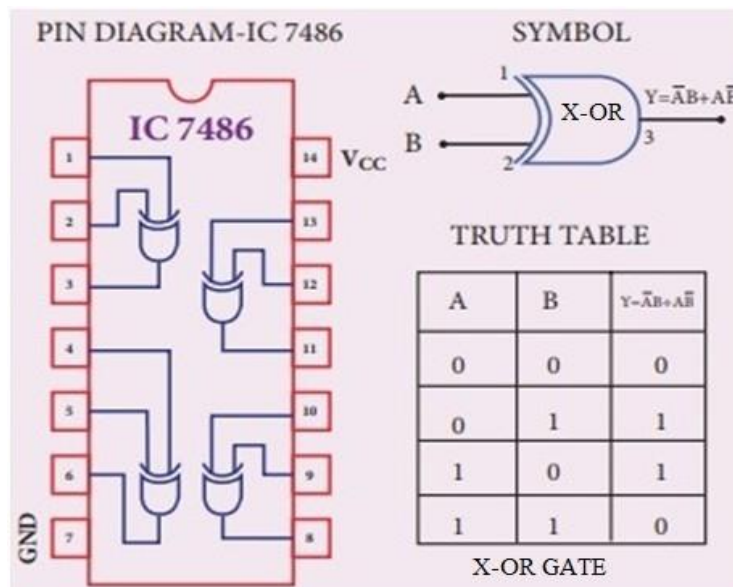


Truth Table of NOR Gate

INPUTs		Theoretical OUTPUT	Output observed
A	B	$Y = \overline{A+B}$	$Y_1 = \overline{A+B}$
0	0	1	
0	1	0	
1	0	0	
1	1	0	

Exclusive OR (X-OR) Gate

X-OR gate produces an output as 1, when number of 1's at its inputs is **odd**, otherwise output is 0. It has two inputs and one output.

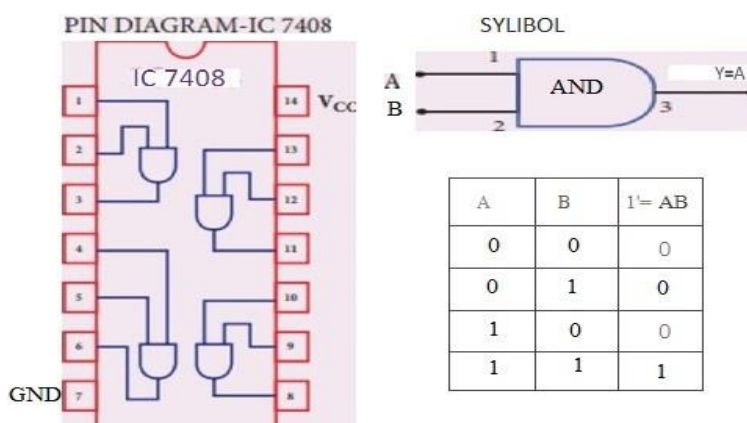


Truth table of Exclusive OR Gate

Inputs		Theoretical O/P	Observed O/P
A	B	$Y = A \oplus B$	$Y = A \oplus B$
0	0	0	
0	1	1	
1	0	1	
1	1	0	

i) AND Gate

AND gate produces an output as 1, when all its inputs are 1; otherwise the output is 0. This gate can have minimum 2 inputs but output is always one. Its output is 0 when any input is 0.

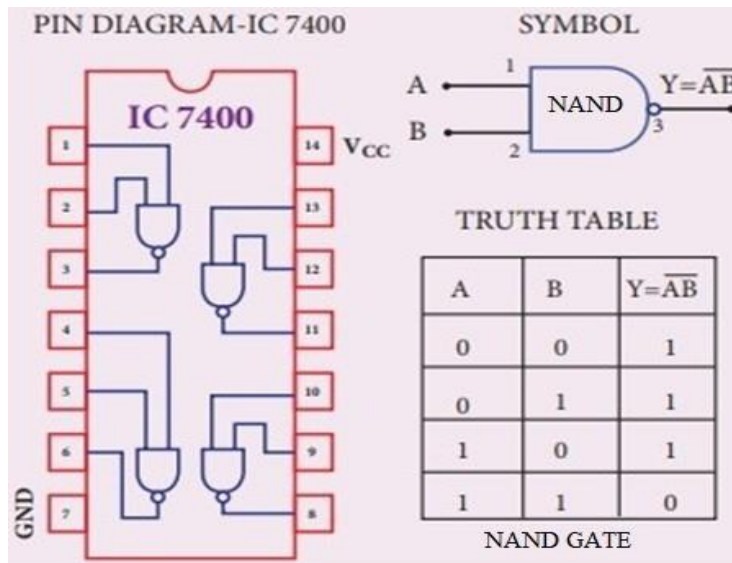


Truth Table of AND Gate

INPUTs		Theoretical OUTPUT	Output observed
A	B	$Y = A.B$	$Y_1 = A.B$
0	0	0	
0	1	0	
1	0	0	
1	1	1	

NAND Gate

NAND gate is actually a series of AND gate with NOT gate. If we connect the output of an AND gate to the input of a NOT gate, this combination will work as NOT-AND or NAND gate. Its output is 1 when any or all inputs are 0, otherwise output is 1.



Truth Table of NAND Gate

INPUTs		Theoretical OUTPUT	Output observed
A	B	$Y = \overline{A.B}$	$Y_1 = \overline{A.B}$
0	0	1	
0	1	1	
1	0	1	
1	1	0	

PROCEDURE

- 1) Write down truth table of the OR Gate as in table 01.
- 2) Fix the OR gate IC 7432 in the Digital Trainer.
- 3) Select any one of the gates in the IC.
- 4) Connect input terminals of the gate to any input points (switches) in the Trainer.
- 5) Connect output terminal of the gate to any output device (LED) in the trainer.
- 6) Connect V_{CC} terminal to +5V and GND terminal to 0V (Biasing supply) in the trainer.
- 7) Switch on the power supply to the trainer.
- 8) Apply inputs to the gate as in the truth table. Observe the output and note down in table 01.
- 9) Verify that observed output is equal to the theoretical output.
- 10) Switch off the power supply. If you like, verify the other gates in the IC.
- 11) Repeat the expt for NOT, NOR, X-OR, AND, NAND, Gate using related and related diagrams.

Precautions :(i) VCC and ground pins must not be interchanged while making connections. Otherwise the chip will be damaged.

(ii) The pin configuration for NOR gate is different from other gates

RESULT: The truth table of logic gates AND, OR, NOR, X-OR, NAND, NOT using integrated circuits is verified.

Signature of staff

Connect and test an IR proximity sensor to a Digital circuit.

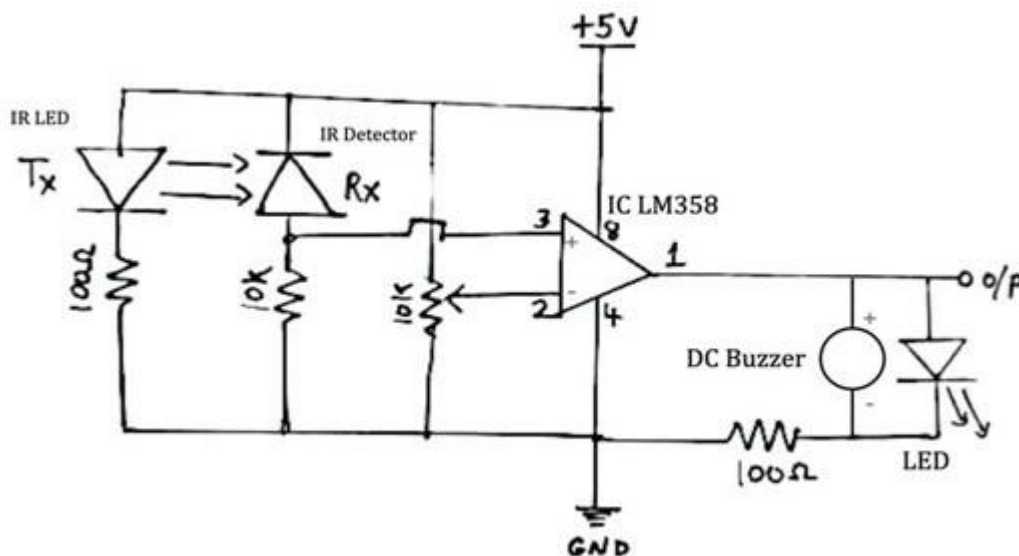
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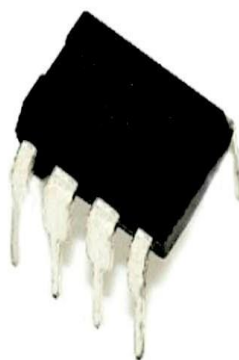
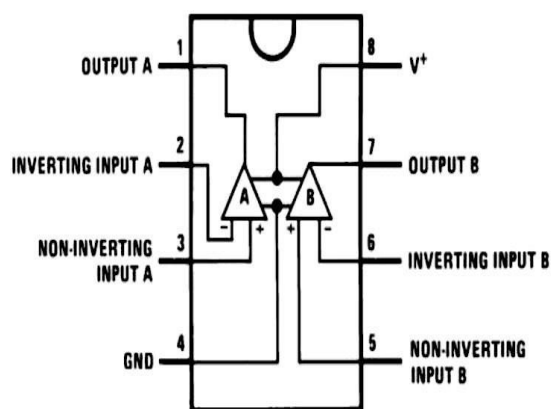
Aim: To connect and test an IR proximity sensor to a digital circuit.

Apparatus required

SLNo.	Item with Specification	Quantity
1	LM 358 IC	1
2	100 Ω resistors	2
3	15 K resistor	1
4	IR LEDs	2
5	LEDs (white)	2
6	Potentiometer 10 K Ω	1

Circuit Diagram





Theory

IR stands for Infrared which is a light with wavelength which is not visible to human eyes but cameras can see it.

It is used in many applications such as TV remote and Night-vision cameras.

In a Proximity Sensor an IR LED and a Photodiode is used to find a obstacles. The IR LED transmits light in forward direction when an obstacle is ahead the light reflects and the Photodiode is activated by this method Obstacle is detected.

Procedure

1. Connections are made as per the circuit diagram.
2. 5V supply is given to the circuit.
3. An obstacle is moved close to the IR LED and sound of buzzer along with glowing of output LED is observed.
4. The sensitivity of the device is varied by varying the trimmer.
5. The supply is switched off.

Result: IR proximity sensor is tested.

Signature of staff

Connect and test a relay circuit using an Optocoupler

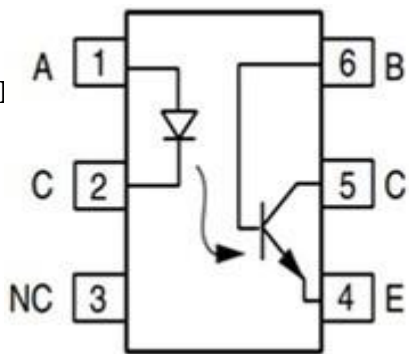
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Aim: To connect and test a relay circuit using an opto coupler.

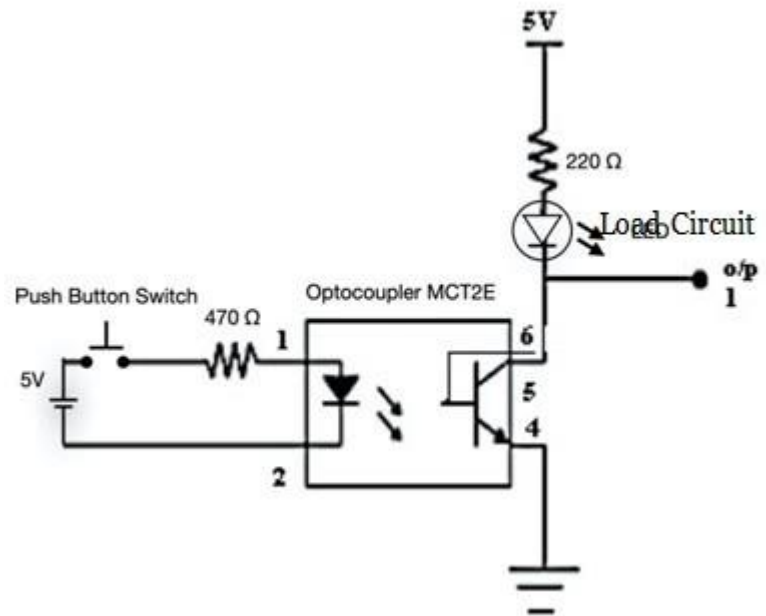
Apparatus required

Sl.No.	Item with Specification	Quantity
1	Optocoupler IC MCT2E	1
2	NPN Transistor SL100	1
3	RPS	1
4	Cell 1.5V	2
5	LED	1
6	220 Ω , 470 Ω Resistors	1 each
7	Regulated Power Supply 0-30V DC	1

Circuit Diagram



onfiguration of MCT2E



Theory:

Opto-isolators or Opto-couplers, are made up of a light emitting device, and a light sensitive device, all wrapped up in one package, but with no electrical connection between the two, just a beam of light. The light emitter is nearly always an LED. The light sensitive device may be a photodiode, phototransistor, or more esoteric devices such as thyristors, TRIACs etc.

The MCT2E series of opto-coupler devices each consist of gallium arsenide infrared LED and a silicon NPN phototransistor. They are packaged in a 6-pin DIP package and available in wide-lead spacing. When light falls on the base-emitter junction then it switches and pin5 goes to zero.

When zero volts is given at the input side LED, then the light doesn't fall on transistor so it doesn't conduct which gives output LED OFF.

When some voltage (3V-5V) is given as input then light falls on photo transistor so that it conducts, that makes photo transistor is switched ON and it forms a short circuit this makes the output LED connected to ground through photo transistor, hence it glows.

Procedure:

1. Connections are made as per the circuit diagram.
2. Supply is given to both control circuit and load circuit by turning on both RPS.
3. Push button switch is pressed and the glow of output LED is observed.
4. The supply is switched off.

Result: Relay circuit using optocoupler is realized.

Signature of staff

(Photo Diode & Transistor)

Identify different application microcontroller.

Date:

Aim: To identify MCS 51 MICROCONTROLLER variants

MCS 51 or 8051 microcontroller is a very popular 8-bit microcontroller introduced by Intel in the year 1981 and it has become almost the academic standard now a days. It is available as a 40-pin DIP chip and works at +5 Volts DC. The salient features of 8051 controller are given below.

SALIENT FEATURES

KB on chip program memory (ROM or EPROM)).

2.128 bytes on chip data memory (RAM).

3.8-bit data bus

4.16-bit address bus

5.32 general purpose registers each of 8bits

6. Two -16 bit timers T0 and T1

7. Four Parallel ports each of 8-bits (PORT0, PORT1, PORT2, PORT3) with a total of 32 /Olines.

8. One 16-bit program counter and One 16-bit DPTR (data pointer)

9. One Microsecond instruction cycle with 12 MHz Crystal.

Variants of MCS 51

Microcontroller	On chip RAM (Bytes)	On chip program memory	Timers/ Counters	Interrupts	Serial ports
8031	128	None	2	5	1
8032	256	None	3	6	1
8051	128	4K ROM	2	5	1
8052	256	8K ROM	3	6	1
8751	128	4K EPROM	2	5	1
8752	256	8K EPROM	3	6	1

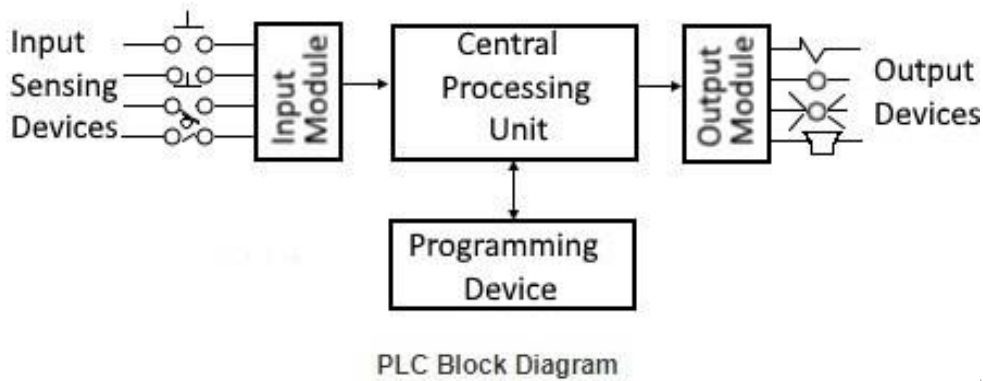
- Identify commercially available PLC and their specifications

Date:

PLC stands for “Programmable Logic Controller”. A PLC is a computer specially designed to operate reliably under harsh industrial environments – such as extreme temperatures, wet, dry, and/or dusty conditions. PLCs are used to automate industrial processes such as a manufacturing plant’s assembly line, an ore processing plant, or a wastewater treatment plant.

Physical Structure of PLC

The structure of a PLC is almost similar to a computer’s architecture.(Fig. Below)



Some of the commercially available PLC includes

1. AllenBradley
2. ABB
3. Siemens
4. MitsubishiPLC
5. HitachiPLC
6. DeltaPLC
7. General Electric (GE) PLC
8. HoneywellPLC

PLC Applications; PLCs have a variety of applications and uses, including

1. Process Automation Plants (e.g. mining, oil & gas)
2. Glass Industry
3. Paper Industry
4. Cement Manufacturing
5. in boilers – Thermal Power Plants

Specifications of PLCs

Total Number of Points:	Number of points is the sum of the inputs and the outputs. PLCs may be specified by any possible combination of these values.
Number of Instructions:	Instructions are standard operations (such as math functions) available to PLC software.
Number of Inputs:	Input signals the PLC can accept.
Number of Outputs:	Output signals the PLC can provide.
Inputs	
DC Input	Discrete DC voltage or current level.
AC Input	Discrete AC voltage or current level.
Analog Input	Variable current or voltage input such as 0-10V, 4-20mA, etc.
Thermocouple Input	Input for a thermocouple or thermocouple series.
RTD Input	Input designed for a resistance temperature detector.
Frequency / Pulse Input	Input from any device that outputs a frequency signal or pulse train.

Transistor Input	Input from a solid-state device such as a sensor; typically includes choice of sinking or sourcing.
Interrupt Input	Input for a digital signal that can stop, resume, or redirect a component of the resident software.
Other Input	Other unlisted or specialized input.
Output	
DC Output	Discrete DC voltage or current level.
AC Output	Discrete AC voltage or current level.
Relay Output	Switch type output; contacts are open or closed.
Analog Output	Variable voltage or current level, e.g., 0-10V, 4-20 <u>mA</u> current loop, etc.
Frequency / Pulse Output	Output is in the form of a frequency or pulse train. Pulses can be used to trigger or control external devices.

Result: The various variants of MCS51 and commercially available PLCs and their specifications and applications are studied.

Signature of staff

COURSE ASSESSMENT AND EVALUATION CHART

Sl.No	Assessment	Duration	Max marks	Conversion	
1.	CIE Assessment 1 (Written Test -1-theory) - At the end of 3 rd week	60 minutes	20	Average of two written tests 20	
2.	CIE Assessment 2 (Written Test -2-theory) - At the end of 13 th week	60 minutes	20		
3.	CIE Assessment 3 (Skill test) - At the end of 5 th week	3 Hours	100	20	Average of three skill tests 20
4.	CIE Assessment 4 (Skill test) - At the end of 7 th week	3 Hours	100		
5.	CIE Assessment 5 (Skill test) - At the end of 9 th week	3 Hours	100		
6.	CIE Assessment 6 (Student activity) - At the end of 11 th week	-	20	20	
7.	Total Continuous Internal Evaluation (CIE) Assessment			60	
8.	Semester End Examination (SEE) Assessment (Practical Test)	3 Hours	100	40	
Total Marks				100	

SCHEME OF VALUATION FOR SKILL TEST (CIE) & SEE

Sl. No.	Particulars	Marks
1.	Identification of meters/ equipment/wires/tools etc.	10
2.	Writing Circuit/writing diagram and Procedure*	25
3.	Conduction	35
4.	Results	10
5.	Viva-voce	20
Total		100

Identification of Meters, Tools, etc.

Some VIVA questions .

- 1) State Ohm's Law. Mention its applications and limitations.
- 2) State Faraday's laws EMI.
- 3) Explain self and mutual induction. Mention their applications.
- 4) Advantages, Disadvantages of electrical energy.
- 5) Applications of electrical energy.
- 6) Motor and generators.
- 7) Define resistance, inductance and capacitance.
- 8) Define frequency, period, rms value, peak value, peak factor, form factor, etc for an alternating quantity.
- 9) Define power and pf.
- 10) Say few words about a p-n junction diode. Mention different types of diode.
- 11) Zener diode as voltage regulator.
- 12) define different Logic gates, write their symbols and truth tables.
- 13) Starters for electric motors.
- 14) Define electric current, voltage, mention their units.
- 15) Define KWH. What is an Energy Meter?
- 16) Compare primary and secondary cells.
- 17) In DC supply- we have +ve, and -ve terminals.
In AC supply- we have phase and neutral terminals.
- 18) There are mainly two types – MC (Moving Coil) and MI (Moving Iron) -
--MC can be used only for DC measurements; MI instruments can be used for both AC and DC measurements.
[But, accuracy of MC instruments is better than MI instruments for DC measurements, hence usually, MC instruments are used to measure DC values, and MI instruments are used to measure AC values.]
- 19) Instrument used to measure voltage is Voltmeter. Instrument used to measure current is Ammeter.
- 20) ---MC instruments have polarities:- That is + or -----MI instruments do not have polarities.
- 20) Instrument used to measure power is watt-meter.
- 21) Instrument used to measure resistance is Ohmmeter.
- 22) Instrument used to measure energy is energymeter.
- 23) Multimeter can measure many different quantities.
- 24) Analog and Digital Meters.
- 25) Single Range and Dual Range meters. Dual Range meters have multiplying factor.
- 26) Multiplying factor of wattmeters.
- 27) UPF (Unity PF) and LPF (Low PF) Wattmeters.
[LPF Watt-meters are used in measurements of low pf loads - like inductive and capacitive.
UPF Wattmeters are used in measurements of HIGH PF loads. - Like resistive/lamp loads.]
- 28) In LPF wattmeters, usually it would be mentioned as "pf = 0.2" on the scale.
- 29) Phase terminal is called as Live wire, live terminal, Hot line, hot wire, line 1, etc.
Neutral wire is usually called as neutral wire or line 2.
- 30) In wiring, - For INDEPENDENT CONTROL, ----the neutral wire will be directly connected to the one end of every load---- The phase wire would be connected to one end of all the switches.

--- The other end of the switch would be connected to the free end of the individual/respective load. When any of the switch is closed, the load is energized as it gets both phase and neutral, and the ckt is closed.
 ----The fuse wire is usually connected in the phase line.----Fuse is an OVER-CURRENT protective device. That is- when current flowing through it exceeds the limit, it burns and breaks the ckt, thereby, current through the load/ckt becomes zero.

31) Staircase wiring

32) In a 3 pin socket, the left terminal/hole is for NEUTRAL, right terminal/hole is for PHASE, and the middle terminal is for the EARTH terminal.

33) Earthing is used to protect you from an electric shock. It does this by providing a path (a protective conductor) for a fault current to flow to earth. It also causes the protective device (either a circuit-breaker or fuse) to switch off the electric current to the circuit that has the fault.

34) Electromagnet. - An electromagnet is a type of magnet in which the magnetic field is produced by an electric current. The magnetic field disappears when the current is turned off.

35) Fleming's Right Hand Rule (for generators) and Left Hand Rule (for motors).

36) Lenz' Law [study only if it is in syllabus].

37) Rheostat - Cut-In and Cut-Out positions.

38) Define magnetic field, flux.

39) Define 1V.

40) Current Coil (CC) and Potential Coils (PC) in Energy Meter and Wattmeters.

41) Integrated Circuits.

42) Fuses

43) Circuit breakers

44) DOL Starter.

